

Visual homogeneity computations in the brain enable solving generic visual tasks

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Georgin Jacob, R. T. Pramod, S. P. Arun 

Department of Electrical Communication Engineering Indian Institute of Science, Bangalore 560012 • Centre for Neuroscience Indian Institute of Science, Bangalore 560012

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Abstract

Many visual tasks involve looking for specific object features. But we also often perform generic tasks where we look for specific property, such as finding an odd item, deciding if two items are same, or if an object has symmetry. How do we solve such tasks? Using well-known principles governing multiple object representations, we show that displays with repeating elements can be distinguished from heterogeneous displays using a property we define as visual homogeneity. In behavior, visual homogeneity predicted response times on visual search and symmetry tasks. Brain imaging during these tasks revealed that visual homogeneity in both tasks is localized to a region in the object-selective cortex. Thus, a novel image property, visual homogeneity, is encoded in a localized brain region, to solve generic visual tasks.

Significance statement

Most visual tasks involve looking for specific features, like finding a face in a crowd. But we often also perform generic tasks where we look for a particular image property – such as finding an odd item, deciding if two items are same, or judging if an object is symmetric. Precisely how we solve such disparate tasks is unclear. Here, we show that these tasks can be solved using a simple property we define as visual homogeneity. In behavior, visual homogeneity predicted response times on visual search and symmetry tasks. In brain imaging, it was localized to a region near the object-selective cortex. Thus, a novel image property, visual homogeneity, is computed by the brain to solve generic visual tasks.

eLife assessment

This study uses carefully designed experiments to generate a **useful** behavioural and neuroimaging dataset on visual cognition. The results provide **solid** evidence for the involvement of higher-order visual cortex in processing visual oddballs and asymmetry. However, the evidence provided for the very strong claims of homogeneity as a novel concept in vision science, separable from existing concepts such as target saliency, is **inadequate**.

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Introduction

Many visual tasks involve looking for specific objects or features, such as a friend in a crowd or selecting vegetables in the market. In such tasks, which have been studied extensively, we form a template in our brain that helps guide eye movements and locate the target (Peelen and Kastner, 2014). This template becomes a decision variable that can be used to solve the task: the degree of match to the template indicates the presence or absence of the desired object. However, we also easily perform tasks that do not involve any specific feature template but rather involve finding a property or relation between items. Examples of these generic tasks include finding an odd item, deciding if two items are same and judging if an object is symmetric. While machine vision algorithms are extremely successful in solving feature-based tasks like object categorization (Serre, 2019), they find generic same-different tasks difficult to solve (Kim et al., 2018; Ricci et al., 2021; Puebla and Bowers, 2022), although recent studies have found certain architectures to generalize quite well (Tartaglioni et al., 2023). How do we perform such tasks with equal ease? What is the underlying template or decision variable that we use to solve generic visual tasks?

To start with, these tasks appear completely different at least in the way we describe them verbally. Even theoretical studies have considered visual search (Verghese, 2001; Wolfe and Horowitz, 2017), same-different judgments (Nickerson, 1969; Petrov, 2009) and symmetry detection (Wagemans, 1997; Bertamini and Makin, 2014) as conceptually separate tasks. However, at a deeper level, these tasks are similar because they all involve discriminating between items with repeating features from those without repeating features. We reasoned that if images with repeating features are somehow represented differently in the brain, this difference could be used to solve all these tasks without requiring separate computations for each task. Here we provide evidence for this hypothesis through behavioural and brain imaging experiments on humans.

Our key predictions are depicted in **Figure 1**. Consider a visual search task where participants have to indicate if a display contains an oddball target (**Figure 1A**) or not (**Figure 1B**). According to the well-known principle of divisive normalization in high-level visual cortex (Zoccolan et al., 2005; Agrawal et al., 2020; Katti and Arun, 2022), the neural response to multiple objects is the average of the single object responses. Accordingly, the response to an array of identical items will be the same as the response to the single item. Moreover, the response to an array containing a target among distractors would lie along the line joining the target and distractor in the (neural) representational space. These possibilities are shown for all possible arrays made from three objects in **Figure 1C**. It can be seen that the homogeneous (target-absent) arrays stand apart since they contain repeating items, whereas the heterogeneous (target-present) arrays come closer since they contain a mixture of items. Since displays with repeating items are further away from the center of this space, this distance can be used to discriminate

them from heterogeneous displays (**Figure 1C**, inset). We note here that this decision process does not capture many other complex forms of visual search but may guide the initial selection process towards possible targets.

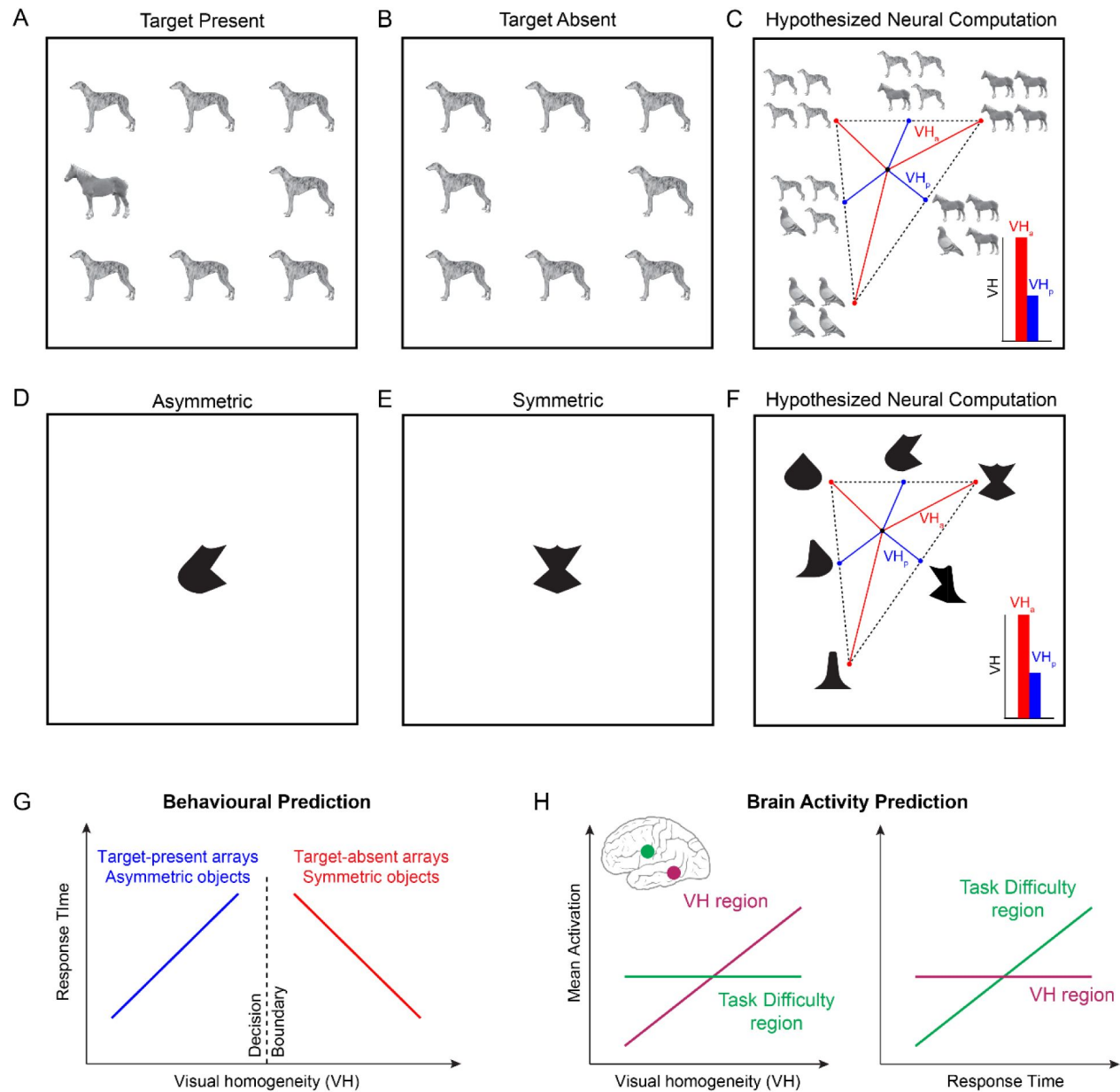


Figure 1.

Solving visual search and symmetry tasks using visual homogeneity.

(A) Example target-present search display, containing a single oddball target (horse) among identical distractors (dog). Participants in such tasks have to indicate whether the display contains an oddball or not, without knowing the features of the target or distractor. This means they have to perform this task by detecting some property of each display rather than some feature contained in it.

(B) Example target-absent search display containing no oddball target.

(C) Hypothesized neural computation for target present/absent judgements. According to multiple object normalization, the response to multiple items is an average of the responses to the individual items. Thus, the response to a target-absent array will be identical to the individual items, whereas the response to a target-present array will lie along the line joining the corresponding target-absent arrays. This causes the target-absent arrays to stay apart (*red lines*), and the target-present arrays to come closer due to mixing (*blue lines*). If we calculate the distance (VH, for visual homogeneity) for each display, then target-absent arrays will have a larger distance to the center (VH_a) compared to target-present arrays (VH_p), and this distance can be used to distinguish between them. *Inset*: Schematic distance from center for target-absent arrays (red) and target-present arrays (blue). Note that this approach might only reflect the initial target selection process involved in oddball visual search but does not capture all forms of visual search.

(D) Example asymmetric object in a symmetry detection task. Here too, participants have to indicate if the display contains a symmetric object or not, without knowing the features of the object itself. This means they have to perform this task by detecting some property in the display.

(E) Example symmetric object in a symmetry detection task.

(F) Hypothesized neural computations for symmetry detection. Following multiple object normalization, the response to an object containing repeated parts is equal the response to the individual part, whereas the response to an object containing two different parts will lie along the line joining the objects with the two parts repeating. This causes symmetric objects to stand apart (*red lines*) and asymmetric objects to come closer due to mixing (*blue lines*). Thus, the visual homogeneity for symmetric objects (VH_s) will be larger than for asymmetric objects (VH_a). *Inset*: Schematic distance from center for symmetric objects (red) and asymmetric objects (blue).

(G) *Behavioral predictions for VH*. If visual homogeneity (VH) is a decision variable in visual search and symmetry detection tasks, then response times (RT) must be largest for displays with VH close to the decision boundary. This predicts opposite correlations between response time and VH for the present/absent or symmetry/asymmetry judgements. It also predicts zero overall correlation between VH and RT.

(H) *Neural predictions for VH*. *Left*: Correlation between brain activations and VH for two hypothetical brain regions. In the VH-encoding region, brain activations should be positively correlated with VH. In any region that encodes task difficulty as indexed by response time, brain activity should show no correlation since VH itself is uncorrelated with RT (see Panel G). *Right*: Correlation between brain activations and RT. Since VH is uncorrelated with RT overall, the region VH should show little or no correlation, whereas the regions encoding task difficulty would show a positive correlation.

We reasoned similarly for symmetry detection: here, participants have to decide if an object is asymmetric (**Figure 1D**) or symmetric (**Figure 1E**). According to multiple object normalization, objects with two different parts would lie along the line joining objects containing the two repeated parts (**Figure 1F**). Indeed, both symmetric and asymmetric objects show part

summation in their neural responses (Pramod and Arun, 2018). Consequently, symmetric objects will be further away from the centre of this space compared to asymmetric objects, and this can be the basis for discriminating them (**Figure 1F**, *inset*).

We define this distance from the center for each image as its *visual homogeneity* (VH). We made two key experimental predictions to test in behavioural and brain imaging experiments. First, if VH is being used to solve visual search and symmetry detection tasks, then responses should be slowest for displays with VH close to the decision boundary and faster for displays with VH far away (**Figure 1G**). This predicts opposite correlations between response time and VH: for target-present arrays and asymmetric objects, the response time should be positively correlated with VH. By contrast, for target-absent arrays and symmetric objects, response time should be negatively correlated with VH. Importantly, because response times can be positively or negatively correlated with VH, the net correlation between response time and VH will be close to zero. Second, if VH is encoded by a dedicated brain region, then brain activity in that region will be positively correlated with VH (**Figure 1H**). Such a positive correlation cannot be explained easily by cognitive processes linked to response time such as attention or task difficulty, since response times will have zero correlation with the mean activity of this region.

Results

In Experiments 1-2, we investigated whether visual homogeneity computations could explain decisions about targets being present or absent in an array. Since visual homogeneity requires measuring distance in perceptual space, we set out to first characterize the underlying representation of a set of natural objects using measurements of perceptual dissimilarity.

Measuring perceptual space for natural objects

In Experiment 1, 16 human participants viewed arrays made from a set of 32 grayscale natural objects, with an oddball on the left or right (**Figure 2A**), and had to indicate the side on which the oddball appeared using a key press. Participants were highly accurate and consistent in their responses during this task (accuracy, mean $\pm$ sd: $98.8 \pm 0.9\%$; correlation between mean response times of even- and odd-numbered participants: $r = 0.91$, $p < 0.0001$ across all $^{32}C_2 = 496$ object pairs). The reciprocal of response time is a measure of perceptual distance (or dissimilarity) between the two images (Arun, 2012). To visualize the underlying object representation, we performed a multidimensional scaling analysis, which embeds objects in a multidimensional space such that their pairwise dissimilarities match the experimentally observed dissimilarities (see Methods). The resulting two-dimensional embedding of all objects is shown in **Figure 2B**. In the resulting plot, nearby objects correspond to hard searches, and far away objects correspond to easy searches. Such representations reconstructed from behavioural data closely match population neural responses in high-level visual areas (Op de Beeck et al., 2001; Sripathi and Olson, 2010). To capture the object representation accurately, we took the multidimensional embedding of all objects and treated the values along each dimension as the responses of an individual artificial neuron. We selected the number of dimensions in the multidimensional embedding so that the correlation between the observed and embedding dissimilarities matches the noise ceiling in the data. Subsequently, we averaged these single object responses to obtain responses to larger visual search arrays, as detailed below.

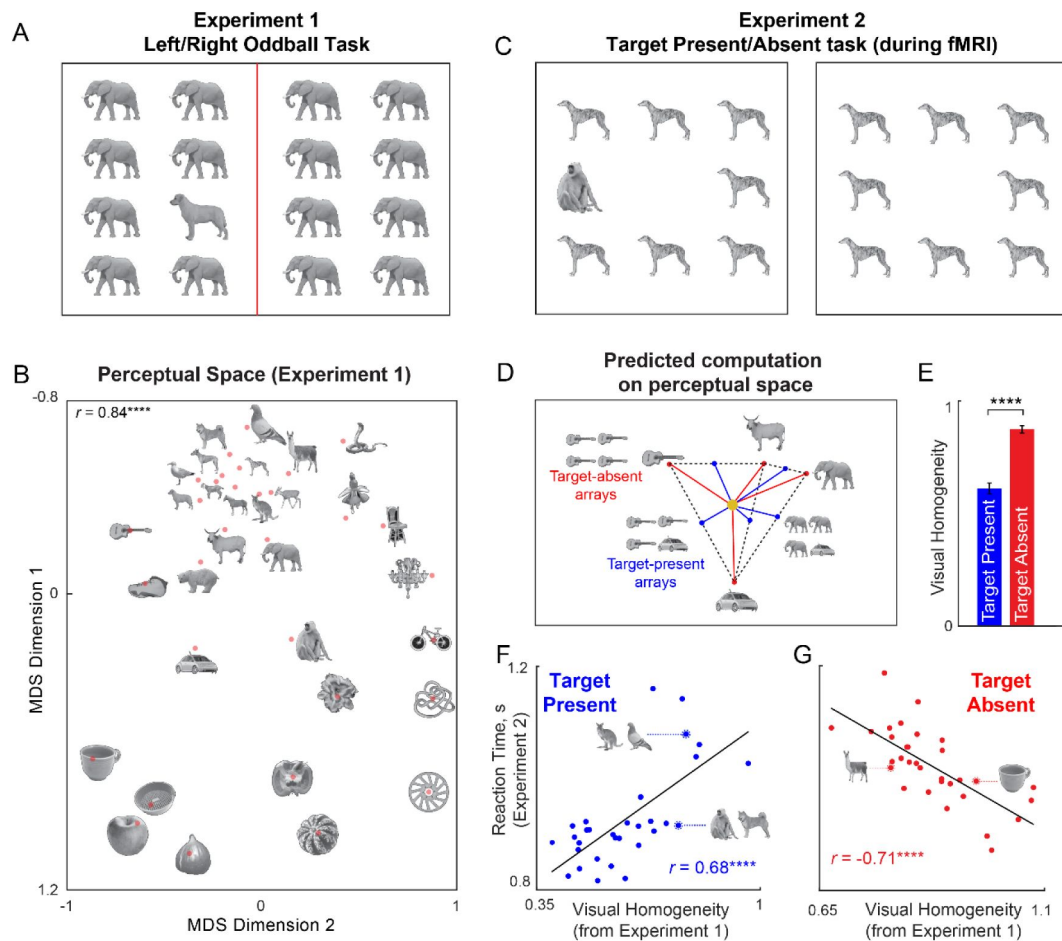


Figure 2.

Visual homogeneity predicts target present/absent responses

(A) Example search array in an oddball search task (Experiment 1). Participants viewed an array containing identical items except for an oddball present either on the left or right side, and had to indicate using a key press which side the oddball appeared. The reciprocal of average search time was taken as the perceptual distance between the target and distractor items. We measured all possible pairwise distances for 32 grayscale natural objects in this manner.

(B) Perceptual space reconstructed using multidimensional scaling performed on the pairwise perceptual dissimilarities. In the resulting plot, nearby objects represent hard searches, and far away objects represent easy searches. Some images are shown at a small size due to space constraints; in the actual experiment, all objects were equated to have the same longer dimension. The correlation on the top right indicates the match between the distances in the 2D plot with the observed pairwise distances (**** is $p < 0.00005$).

(C) Example display from Experiment 2. Participants performed this task inside the scanner. On each trial, they had to indicate whether an oddball target is present or absent using a key press.

(D) Predicted response to target-present and target-absent arrays, using the principle that the neural response to multiple items is the average of the individual item responses. This predicts that target-present arrays become similar due to mixing of responses, whereas target-absent arrays stand apart. Consequently, these two types of displays can be distinguished using their distance to a central point in this space. We define this distance as visual homogeneity, and it is obtained by finding the optimum center that maximizes the difference in correlations with response times (see Methods).

(E) Mean visual homogeneity relative to the optimum center for target-present and target-absent displays. Error bars represent s.e.m across all displays. Asterisks represent statistical significance (**** is $p < 0.00005$, unpaired rank-sum test comparing visual homogeneity for 32 target-absent and 32 target-present arrays).

(F) Response time for target-present searches in Experiment 2 plotted against visual homogeneity calculated from Experiment 1. Asterisks represent statistical significance of the correlation (**** is $p < 0.00005$). Note that a single model is fit to find the optimum center in representational space that predicts the response times for both target-present and target-absent searches.

(G) Response time for target-absent searches in Experiment 2 plotted against visual homogeneity calculated from Experiment 1. Asterisks represent statistical significance of the correlation (**** is $p < 0.00005$).

Visual homogeneity predicts target present/absent judgments (Experiments 1-2)

Having characterized the underlying perceptual representation for single objects, we set out to investigate whether target present/absent responses during visual search can be explained using this representation. In Experiment 2, 16 human participants viewed an array of items on each trial, and indicated using a key press whether there was an oddball target present or not (**Figure 2C**). This task was performed inside an MRI scanner to simultaneously observe both brain activity and behaviour. Participants were highly accurate and consistent in their responses (accuracy, mean $\pm$ sd: $95 \pm 3\%$; correlation between average response times of even- and odd-numbered participants: $r = 0.86$, $p < 0.0001$ across 32 target-present searches, $r = 0.63$, $p < 0.001$ across 32 target-absent searches).

Next we set out to predict the responses to target-present and target-absent search displays containing these objects. We first took the object coordinates returned by multidimensional scaling in Experiment 1 as neural responses of multiple neurons. We then used a well-known principle of object representations in high-level visual areas: the response to multiple objects is the average of the single object responses (Zoccolan et al., 2005; Agrawal et al., 2020). Thus, we took the response vector for a target-present array to be the average of the response vectors of the target and distractor (**Figure 2D**). Likewise, we took the response vector for a target-absent array to be equal to the response vector of the single item. We then asked if there is any point in this multidimensional representation such that distances from this point to the target-present and target-absent response vectors can accurately predict the target-present and target-absent response times with a positive and negative correlation respectively (see Methods). Note that this model has as many free parameters as the coordinates of this unknown point or center in multidimensional space, and this model can simultaneously predict both target-present and target-absent judgments. We used nonlinear optimization to find the coordinates of the center to best match the data (see Methods).

We denoted the distance of each display to the optimized center as the visual homogeneity. As expected, the visual homogeneity of target-present arrays was significantly smaller than target-absent arrays (**Figure 2E**). The resulting model predictions are shown in **Figure 2F-G**. The response times for target-present searches were positively correlated with visual homogeneity ($r = 0.68$, $p < 0.0001$; **Figure 2F**). By contrast, the response times for target-absent searches were negatively correlated with visual homogeneity ($r = -0.71$, $p < 0.0001$; **Figure 2G**). This is exactly as predicted if visual homogeneity is the underlying decision variable (**Figure 1G**). We note that the range of visual homogeneity values for target-present and target-absent searches do overlap, suggesting that visual homogeneity contributes but does not fully determine task performance. Rather, we suggest that visual homogeneity provides a useful and initial first guess at the presence or absence of a target, which can be refined further through detailed scrutiny.

To confirm that the above model fits are not due to overfitting, we performed a leave-one-out cross validation analysis, where we left out all target-present and target-absent searches involving a particular image, and then predicted these searches by calculating visual homogeneity. This too yielded similar positive and negative correlations ($r = 0.63$, $p < 0.0001$ for target-present, $r = -0.63$, $p < 0.001$ for target-absent).

These findings are non-trivial for several reasons. First, we have shown that a specific decision variable, computed over an underlying object representation, can be used to make target present/absent judgements, without necessarily knowing the precise features of the target or distractor. Second, we have identified a specific image property that explains why target-absent response times vary so systematically. If target-distractor dissimilarity were the sole determining

factor in visual search, it would predict no systematic variation in target-absent searches since the target-distractor dissimilarity is zero. Our results elucidate this puzzle by showing that visual homogeneity varies systematically across images, and that this explains systematic variation in target-absent response times. To the best of our knowledge, our model provides for the first time a unified explanation to explain how target present/absent judgements might be made in a generic visual search task.

We performed several additional analyses to validate these results and confirm their generality. First, if heterogeneous displays elicit similar neural responses due to mixing, then their average distance to other objects must be related to their visual homogeneity. We confirmed that this was indeed the case, suggesting that the average distance of an object from all other objects in visual search can predict visual homogeneity (Section S1). Second, the above analysis was based on taking the neural response to oddball arrays to be the average of the target and distractor responses. To confirm that averaging was indeed optimal, we repeated the above analysis by assuming a range of relative weights between the target and distractor. The best correlation was obtained for almost equal weights in the lateral occipital (LO) region, consistent with averaging and its role in the underlying perceptual representation (Section S1). Third, we performed several additional experiments on a larger set of natural objects as well as on silhouette shapes. In all cases, present/absent responses were explained using visual homogeneity (Section S2).

In sum, we conclude that visual homogeneity can explain oddball target present/absent judgements during visual search.

Visual homogeneity predicts same/different responses

We have proposed that visual homogeneity can be used to solve any task that requires discriminating between homogeneous and heterogeneous displays. In Experiments 1-2, we have shown that visual homogeneity predicts target present/absent responses in visual search. We performed an additional experiment to assess whether visual homogeneity can be used to solve an entirely different task, namely a same-different task. In this task, participants have to indicate whether two items are the same or different. We note that instructions to participants for the same/different task ("you have to indicate if the two items are same or different") are quite different from the visual search task ("you have to indicate whether an oddball target is present or absent"). Yet both tasks involve discriminating between homogeneous and heterogeneous displays. We therefore predicted that "same" responses would be correlated with target-absent judgements and "different" responses would be correlated with target-present judgements. Remarkably, this was indeed the case (Section S3), demonstrating that same/different responses can also be predicted using visual homogeneity.

Visual homogeneity is independent of experimental context

In the above analyses, visual homogeneity was calculated for each display as its distance from an optimum center in perceptual space. This raises the possibility that visual homogeneity could be modified depending on experimental context since it could depend on the set of objects relative to which the visual homogeneity is computed. We performed a number of experiments to evaluate this possibility: we found that target-absent response times, which index visual homogeneity, are unaffected by a variety of experimental context manipulations (Section S4).

We therefore propose that visual homogeneity is an image-computable property that remains stable across tasks. Moreover, it can be empirically estimated as the reciprocal of the target-absent response times in a visual search task.

A localized brain region encodes visual homogeneity (Experiment 2)

So far, we have found that target present/absent response times had opposite correlations with visual homogeneity ([Figure 2F-G](#)), suggesting that visual homogeneity is a possible decision variable for this task. Therefore, we reasoned that visual homogeneity may be localized to specific brain regions, such as in the visual or prefrontal cortices. Since the task in Experiment 2 was performed by participants inside an MRI scanner, we set out to investigate this issue by analyzing their brain activations.

We estimated brain activations in each voxel for individual target-present and target-absent search arrays (see Methods). To identify the brain regions whose activations correlated with visual homogeneity, we performed a whole-brain searchlight analysis. For each voxel, we calculated the mean activity in a 3x3x3 volume centered on that voxel (averaged across voxels and participants) for each present/absent search display, and calculated its correlation with visual homogeneity predictions derived from behavior (see Methods). The resulting map is shown in [Figure 3A](#). Visual homogeneity was encoded in a highly localized region just anterior of the lateral occipital (LO) region, with additional weak activations in the parietal and frontal regions. To compare these trends across key visual regions, we calculated the correlation between mean activation and visual homogeneity for each region. This revealed visual homogeneity to be encoded strongly in this region VH, and only weakly in other visual regions ([Figure 3D](#)).

To ensure that the high match between visual homogeneity and neural activations in the VH region is not due to an artefact of voxel selection, we performed subject-level analysis (Section S5). We repeated the searchlight analysis for each subject and defined VH region for each subject. We find this VH region consistently anterior to the LO region in each subject. Next, we divided participants into two groups, and repeated the brain-wide searchlight analysis. Importantly, the match between mean activation and visual homogeneity remained significant even when the VH region was defined using one group of participants and the correlation was calculated using the mean activations of the other group (Section S5).

To confirm that neural activations in VH region are not driven by other cognitive processes linked to response time, such as attention, we performed a whole-brain searchlight analysis using response times across both target-present and target-absent searches. Proceeding as before, we calculated the correlation between mean activations to the target-present, target-absent and all displays with the respective response times. The resulting maps show that mean activations in the VH region are uncorrelated with response times overall (Section S5). By contrast, activations in EVC and LO are negatively correlated with response times, suggesting that faster responses are driven by higher activation of these areas. Finally, mean activation of parietal and prefrontal regions were strongly correlated with response times, consistent with their role in attentional modulation (Section S5).

Object representations in LO match with visual search dissimilarities

To investigate the neural space on which visual homogeneity is being computed, we performed a dissimilarity analysis. Since target-absent displays contain multiple instances of a single item, we took the neural response to target-absent displays as a proxy for the response to single items. For each pair of objects, we took the neural activations in a 3x3x3 neighborhood centered around a given voxel and calculated the Euclidean distance between the two 27-dimensional response vectors (averaged across participants). In this manner, we calculated the neural dissimilarity for all $^{32}C_2 = 496$ pairs of objects used in the experiment, and calculated the correlation between the neural dissimilarity in each local neighborhood and the perceptual dissimilarities for the same

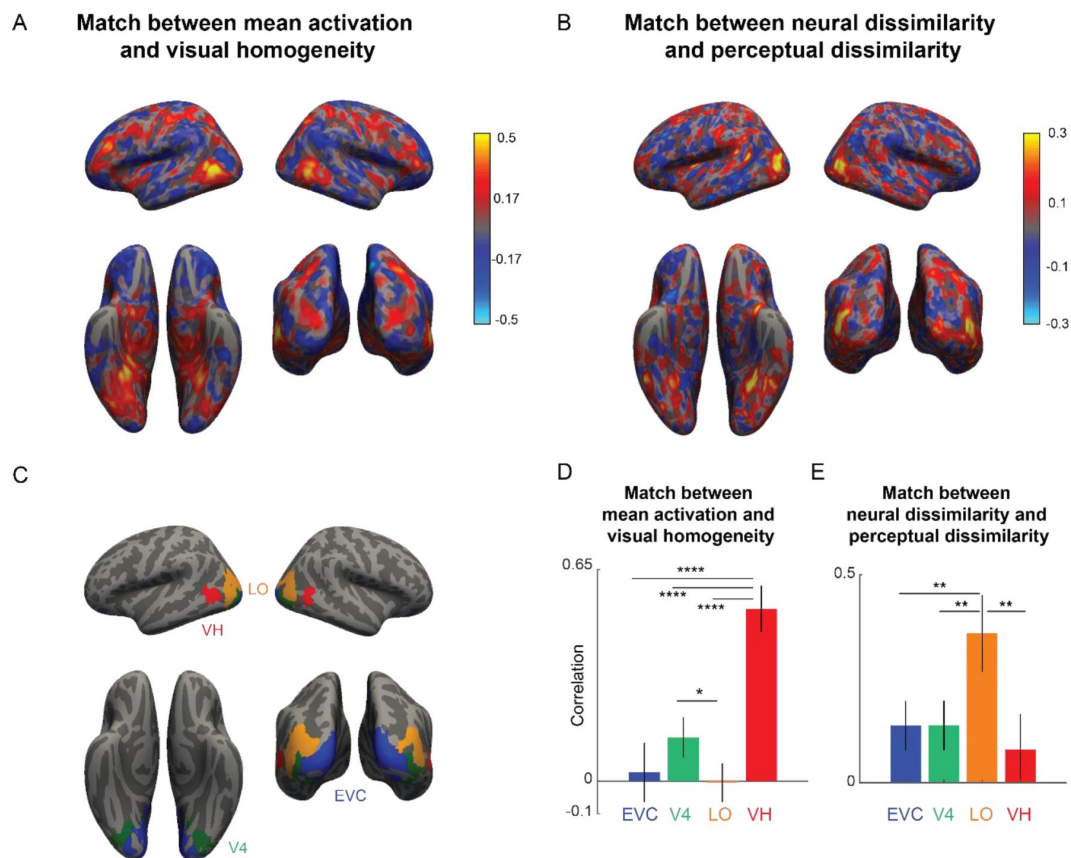


Figure 3

A localized brain region encodes visual homogeneity

A. Searchlight map showing the correlation between mean activation in each 3x3x3 voxel neighborhood and visual homogeneity.

B. Searchlight map showing the correlation between neural dissimilarity in each 3x3x3 voxel neighborhood and perceptual dissimilarity measured in visual search.

C. Key visual regions identified using standard anatomical masks: early visual cortex (EVC), area V4, lateral occipital (LO) region. The visual homogeneity (VH) region was identified using the searchlight map in Panel A.

D. Correlation between the mean activation and visual homogeneity in key visual regions EVC, V4, LO and VH. Error bars represent standard deviation of the correlation obtained using a bootstrap process, by repeatedly sampling participants with replacement for 10,000 times. Asterisks represent statistical significance, estimated by calculating the fraction of bootstrap samples in which the observed trend was violated (* is $p < 0.05$, ** is $p < 0.01$, **** is $p < 0.0001$).

E. Correlation between neural dissimilarity in key visual regions with perceptual dissimilarity. Error bars represent the standard deviation of correlation obtained using a bootstrap process, by repeatedly sampling participants with replacement 10,000 times. Asterisks represent statistical significance, estimated by calculating the fraction of bootstrap samples in which the observed trend was violated (** is $p < 0.001$).

objects measured using oddball search in Experiment 1. The resulting map is shown in **Figure 3B** [↗](#). It can be seen that perceptual dissimilarities from visual search are best correlated in the lateral occipital region, consistent with previous studies (**Figure 3E** [↗](#)). To compare these trends across key visual regions, we performed this analysis for early visual cortex (EVC), area V4, LO and for the newly identified region VH (average MNI coordinates (x, y, z): (-48, -59, -6) with 111 voxels in the left hemisphere; (49, -56, -7) with 60 voxels in the right hemisphere). Perceptual dissimilarities matched best with neural dissimilarities in LO compared to the other visual regions (**Figure 3E** [↗](#)). We conclude that neural representations in LO match with perceptual space. This is concordant with many previous studies (Haushofer et al., 2008 [↗](#); Kriegeskorte et al., 2008 [↗](#); Agrawal et al., 2020 [↗](#); Storrs et al., 2021 [↗](#); Ayzenberg et al., 2022 [↗](#)).

Equal weights for target and distractor in target-present array responses

In the preceding sections, visual homogeneity was calculated using behavioural experiments assuming a neural representation that gives equal weights to the target and distractor. We tested this assumption experimentally by asking whether neural responses to target-present displays can be predicted using the response to the target and distractor items separately. The resulting maps revealed that target-present arrays were accurately predicted as a linear sum of the constituent items, with roughly equal weights for the target and distractor (Section S5).

Visual homogeneity predicts symmetry perception (Experiments 3-4)

The preceding sections show that visual homogeneity predicts target present/absent responses as well same/different responses. We have proposed that visual homogeneity can be used to solve any task that involves discriminating homogeneous and heterogeneous displays. In Experiments 3 & 4, we extend the generality of these findings to an entirely different task, namely symmetry perception. Here, asymmetric objects are akin to heterogeneous displays whereas symmetric objects are like homogeneous displays.

In Experiment 3, we measured perceptual dissimilarities for a set of 64 objects (32 symmetric, 32 asymmetric objects) made from a common set of parts. On each trial, participants viewed a search array with identical items except for one oddball, and had to indicate the side (left/right) on which the oddball appeared using a key press. An example search array is shown in **Figure 4A** [↗](#). Participants performed searches involving all possible ${}^{64}C_2 = 2,016$ pairs of objects. Participants made highly accurate and consistent responses on this task (accuracy, mean $\pm$ sd: $98.5 \pm 1.33\%$; correlation between average response times from even- and odd-numbered subjects: $r = 0.88$, $p < 0.0001$ across 2,016 searches). As before, we took the perceptual dissimilarity between each pair of objects to be the reciprocal of the average response time for displays with either item as target and the other as distractor. To visualize the underlying object representation, we performed a multidimensional scaling analysis, which embeds objects in a multidimensional space such that their pairwise dissimilarities match the experimentally observed dissimilarities. The resulting plot for two dimensions is shown in **Figure 4B** [↗](#), where nearby objects correspond to similar searches. It can be seen that symmetric objects are generally more spread apart than asymmetric objects, suggesting that visual homogeneity could discriminate between symmetric and asymmetric objects.

In Experiment 4, we tested this prediction experimentally using a symmetry detection task that was performed by participants inside an MRI scanner. On each trial, participants viewed a briefly presented object, and had to indicate whether the object was symmetric or asymmetric using a key press (**Figure 4C** [↗](#)). Participants made accurate and consistent responses in this task (accuracy, mean $\pm$ sd: $97.7 \pm 1.7\%$; correlation between response times of odd- and even-numbered participants: $r = 0.47$, $p < 0.0001$).

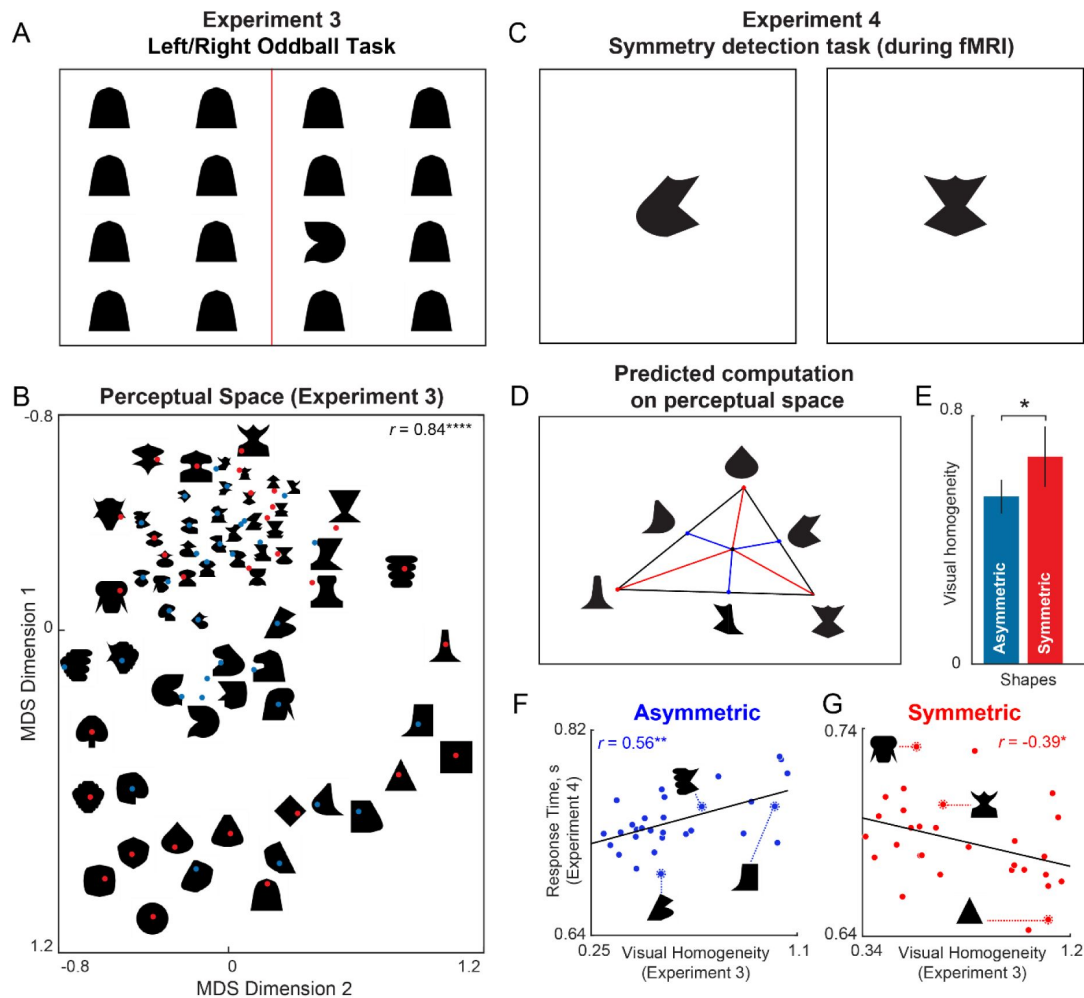


Figure 4.

Visual homogeneity predicts symmetry perception

(A) Example search array in Experiment 3. Participants viewed an array containing identical items except for an oddball present either on the left or right side, and had to indicate using a key press which side the oddball appeared. The reciprocal of average search time was taken as the perceptual distance between the target and distractor items. We measured all possible pairwise distances for 64 objects (32 symmetric, 32 asymmetric) in this manner.

(B) Perceptual space reconstructed using multidimensional scaling performed on the pairwise perceptual dissimilarities. In the resulting plot, nearby objects represent hard searches, and far away objects represent easy searches. Some images are shown at a small size due to space constraints; in the actual experiment, all objects were equated to have the same longer dimension. The correlation on the *top right* indicates the match between the distances in the 2D plot with the observed pairwise distances (**** is $p < 0.00005$).

(C) Two example displays from Experiment 4. Participants had to indicate whether the object is symmetric or asymmetric using a key press.

(D) Using the perceptual representation of symmetric and asymmetric objects from Experiment 3, we reasoned that they can be distinguished using their distance to a center in perceptual space. The coordinates of this center are optimized to maximize the match to the observed symmetry detection times.

(E) Visual homogeneity relative to the optimum center for asymmetric and symmetric objects. Error bar represents s.e.m. across images. Asterisks represent statistical significance (* is $p < 0.05$, unpaired rank-sum test comparing visual homogeneity for 32 symmetric and 32 asymmetric objects).

(F) Response time for asymmetric objects in Experiment 4 plotted against visual homogeneity calculated from Experiment 3. Asterisks represent statistical significance of the correlation (** is $p < 0.001$).

(G) Response time for symmetric objects in Experiment 4 plotted against visual homogeneity calculated from Experiment 3. Asterisks represent statistical significance of the correlation (* is $p < 0.05$).

To investigate whether visual homogeneity can be used to predict symmetry judgements, we took the embedding of all objects from Experiment 3, and asked whether there was a center in this multidimensional space such that the distance of each object to this center would be oppositely correlated with response times for symmetric and asymmetric objects (see Methods). The resulting model predictions are shown in **Figure 4E-G**. As predicted, visual homogeneity was significantly larger for symmetric compared to asymmetric objects (visual homogeneity, mean $\pm$ sd: 0.60 ± 0.24 s⁻¹ for asymmetric objects; 0.76 ± 0.29 s⁻¹ for symmetric objects; $p < 0.05$, rank-sum test; **Figure 4E**). For asymmetric objects, symmetry detection response times were positively correlated with visual homogeneity ($r = 0.56$, $p < 0.001$; **Figure 4F**). By contrast, for symmetric objects, response times were negatively correlated with visual homogeneity ($r = -0.39$, $p < 0.05$; **Figure 4G**). These patterns are exactly as expected if visual homogeneity was the underlying decision variable for symmetry detection. However, we note that the range of visual homogeneity values for asymmetric and symmetric objects do overlap, suggesting that visual homogeneity contributes but does not fully determine task performance. We therefore propose that visual homogeneity provides a useful and initial first guess at symmetry in an image, which can be refined further through detailed scrutiny.

To confirm that the above model fits are not due to overfitting, we performed a leave-one-out cross validation analysis, where we left out one object at a time, and then calculated its visual homogeneity. This too yielded similar correlations ($r = 0.44$ for asymmetric, $r = -0.39$ for symmetric objects, $p < 0.05$ in both cases).

In sum, we conclude that visual homogeneity can predict symmetry perception. Taken together, these experiments demonstrate that the same computation (distance from a center) explains two disparate generic visual tasks: symmetry perception and visual search.

Visual homogeneity is encoded by the VH region during symmetry detection

If visual homogeneity is a decision variable for symmetry detection, it could be localized to specific regions in the brain. To investigate this issue, we analyzed the brain activations of participants in Experiment 4.

To investigate the neural substrates of visual homogeneity, we performed a searchlight analysis. For each voxel, we calculated the correlation between mean activations in a 3x3x3 voxel neighborhood and visual homogeneity. This revealed a localized region in the visual cortex as well as some parietal regions where this correlation attained a maximum (**Figure 5A**). This VH region (average MNI coordinates (x, y, z): (-57, -56, -8) with 93 voxels in the left hemisphere; (58, -50, -8) with 73 voxels in the right hemisphere) overlaps with VH region defined during visual search in Experiment 3 (for a detailed comparison, see Section S7).

We note that it is not straightforward to interpret the overlap between the VH regions identified in Experiments 2 & 4, since images in this experiment (Experiment 4) were presented at the fovea, whereas the visual search arrays in Experiment 2 contained items exclusively in the periphery. These experiments were also performed on different sets of participants, so the differences in overlap could be due to individual variability. Indeed, considerable individual variability has been observed in the location of category-selective regions such as FFA (Weiner and Grill-Spector, 2012) and VWFA (Glezer and Riesenhuber, 2013). We propose that testing the same participants on both search and symmetry tasks would reveal VH regions with even larger overlap.

To confirm that neural activations in VH region are not driven by other cognitive processes linked to response time, such as attention, we performed a whole-brain searchlight analysis using response times across both symmetric and asymmetric objects. This revealed that mean

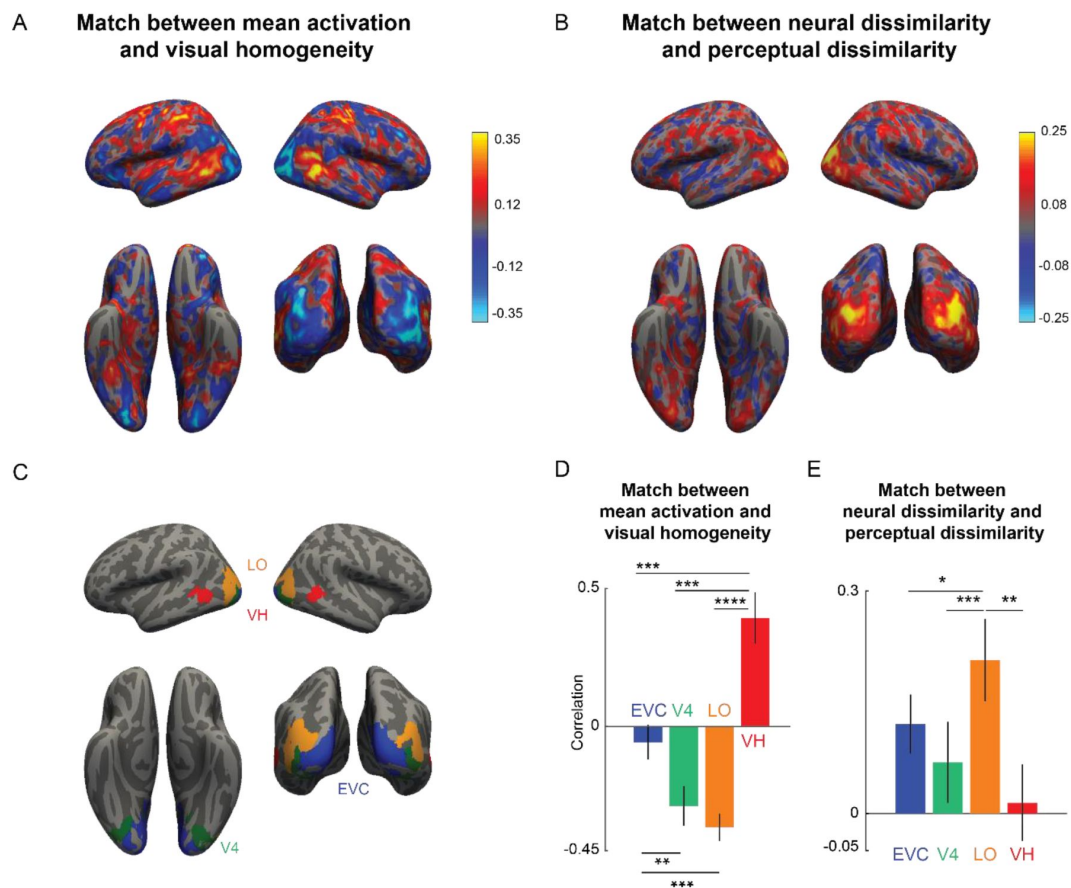


Figure 5

Brain region encoding visual homogeneity during symmetry detection

- (A) Searchlight map showing the correlation between mean activation in each 3x3x3 voxel neighborhood and visual homogeneity.
- (B) Searchlight map showing the correlation between neural dissimilarity in each 3x3x3 voxel neighborhood and perceptual dissimilarity measured in visual search.
- (C) Key visual regions identified using standard anatomical masks: early visual cortex (EVC), area V4, Lateral occipital (LO) region. The visual homogeneity (VH) region was identified using searchlight map in Panel A.
- (D) Correlation between the mean activation and visual homogeneity in key visual regions EVC, V4, LO and VH. Error bars represent standard deviation of the correlation obtained using a bootstrap process, by repeatedly sampling participants with replacement for 10,000 times. Asterisks represent statistical significance, estimated by calculating the fraction of bootstrap samples in which the observed trend was violated (* is $p < 0.05$, ** is $p < 0.01$, **** is $p < 0.0001$).
- (E) Correlation between neural dissimilarity in key visual regions with perceptual dissimilarity. Error bars represent the standard deviation of correlation obtained using a bootstrap process, by repeatedly sampling participants with replacement 10,000 times. Asterisks represent statistical significance, estimated by calculating the fraction of bootstrap samples in which the observed trend was violated (** is $p < 0.001$).

activations in the VH region were poorly correlated with response times overall, whereas other parietal and prefrontal regions strongly correlated with response times consistent with their role in attentional modulation (Section S6).

To investigate the perceptual representation that is being used for visual homogeneity computations, we performed a neural dissimilarity analysis. For each pair of objects, we took the neural activations in a 3x3x3 neighborhood centered around a given voxel and calculated the Euclidean distance between the two 27-dimensional response vectors. In this manner, we calculated the neural dissimilarity for all ${}^{64}C_2 = 2,016$ pairs of objects used in the experiment, and calculated the correlation between the neural dissimilarity in each local neighborhood and the perceptual dissimilarities for the same objects measured using oddball search in Experiment 3. The resulting map is shown in **Figure 5B** [↗](#). The match between neural and perceptual dissimilarity was maximum in the lateral occipital region (**Figure 5B** [↗](#)).

To compare these trends for key visual regions, we repeated this analysis for anatomically defined regions of interest in the visual cortex: early visual cortex (EVC), area V4, the lateral occipital (LO) region, and the VH region defined based on the searchlight map in **Figure 5A** [↗](#). These regions are depicted in **Figure 5C** [↗](#). We then asked how mean activations in each of these regions is correlated with visual homogeneity. This revealed that the match with visual homogeneity is best in the VH region compared to the other regions (**Figure 5D** [↗](#)). To further confirm that visual homogeneity is encoded in a localized region in the symmetry task, we repeated the searchlight analysis on two independent subgroups of participants. This revealed highly similar regions in both groups (Section S6).

Finally, we compared neural dissimilarities and perceptual dissimilarities in each region as before. This revealed that perceptual dissimilarities (measured from Experiment 3, during visual search) matched best with the LO region (**Figure 5E** [↗](#)), suggesting that object representations in LO are the basis for visual homogeneity computations in the VH region.

In sum, our results suggest that visual homogeneity is encoded by the VH region, using object representations present in the adjoining LO region.

Target-absent responses predict symmetry detection

So far, we have shown that visual homogeneity predicts target present/absent responses in visual search as well as symmetry detection responses. These results suggest a direct empirical link between these two tasks. Specifically, since target-absent response time is inversely correlated with visual homogeneity, we can take its reciprocal as an estimate of visual homogeneity. This in turn predicts opposite correlations between symmetry detection times and reciprocal of target-absent response time: in other words, we should see a positive correlation for asymmetric objects, and a negative correlation for symmetric objects. We confirmed these predictions using additional experiments (Section S8). These results reconfirm that a common decision variable, visual homogeneity, drives both target present/absent and symmetry judgements.

Visual homogeneity explains animate categorization

Since visual homogeneity is always calculated relative to a location in perceptual space, we reasoned that shifting this center towards a particular object category would make it a decision variable for object categorization. To test this prediction, we reanalyzed data from a previous study in which participants had to categorize images as belonging to three hierarchical categories: animals, dogs or Labradors (Mohan and Arun, 2012 [↗](#)). By adjusting the center of the perceptual space measured using visual search, we were able to predict categorization responses for all three categories (Section S9). We further reasoned that, if the optimum center for animal/dog/Labrador categorization is close to the default center in perceptual space that predicts target present/absent judgements, then even the default visual homogeneity, as indexed by the reciprocal of target-

absent search time, should predict categorization responses. Interestingly, this was indeed the case (Section S9). We conclude that, at least for the categories tested, visual homogeneity computations can serve as a decision variable for object categorization.

Discussion

Here, we investigated three disparate visual tasks: detecting whether an oddball is present in a search array, deciding if two items are same or different, and judging whether an object is symmetric/asymmetric. Although these tasks are superficially quite different, they all involve discriminating between homogeneous and heterogeneous displays. We defined a new image property computable from the underlying perceptual representation, namely visual homogeneity, that can be used to solve these tasks. We found that visual homogeneity can predict response times in all three tasks. Finally we have found that visual homogeneity estimated from behavior is correlated with mean activations anterior to the lateral occipital region in both visual search and symmetry tasks. This is indicative but not conclusive evidence for a specific function to this part of the high-level visual cortex. Below we discuss these findings in relation to the existing literature.

Visual homogeneity unifies visual search, same-different and symmetry tasks

Our main finding, that a single decision variable (visual homogeneity) can be used to solve three disparate visual tasks (visual search, same/different and symmetry detection) is novel to the best of our knowledge. This finding is interesting and important because it establishes a close correspondence between all three tasks, and explains some unresolved puzzles in each of these tasks, as detailed below.

First, with regards to visual search, theoretical accounts of search are based on signal detection theory (Vergheze, 2001 [↗](#); Wolfe and Horowitz, 2017 [↗](#)), but define the signal only for specific target-distractor pairs. By contrast, the task of detecting whether an oddball item is present requires a more general decision rule that has not been identified. Our results suggest that visual homogeneity is the underlying decision variable. In visual search, target-present search times are widely believed to be driven by target-distractor similarity (Duncan and Humphreys, 1989 [↗](#); Wolfe and Horowitz, 2004 [↗](#)). But target-absent search times also vary systematically for reasons that have not been elucidated previously. The slope of target-absent search times as a function of set size are typically twice the slope of target present searches (Wolfe, 1998 [↗](#)). However this observation is based on averaging across many target-present searches. Since there is only one unique item in a target-absent search array, there must be some image property that causes systematic variations. Our results resolve this puzzle by showing that this systematic variation is driven by visual homogeneity. Finally, our findings also help explain why we sometimes know a target is present without knowing its exact location – this is because the underlying decision variable, visual homogeneity, arises in high-level visual areas with relatively coarse spatial information. However, we note that visual homogeneity computations described in this study do not completely explain all the variance observed in oddball search times in our study – rather they offer an important quantitative model that could explain the initial phase of target selection. We speculate that this initial phase could be shared by all forms of visual search (e.g. searching among non-identical distractors, memory-guided search, conjunction search), and these would be interesting possibilities for future work.

Second, with regards to same-different tasks, most theoretical accounts use signal detection theory but usually with reference to specific stimulus pairs (Nickerson, 1969 [↗](#); Petrov, 2009 [↗](#)). It has long been observed that "different" responses become faster with increasing target-distractor dissimilarity but this trend logically predicts that "same" responses, which have zero difference, should be the slowest (Nickerson, 1967 [↗](#), 1969 [↗](#)). But in fact, "same" responses are faster than

"different" responses. This puzzle has been resolved by assuming a separate decision rule for "same" judgements, making the overall decision process more complex (Petrov, 2009 [↗](#); Goulet, 2020 [↗](#)). Our findings resolve this puzzle by identifying a novel variable, visual homogeneity, which can be used to implement a simple decision rule for making same/different responses. Our findings also explain why some images elicit faster "same" responses than others: this is due to image-to-image differences in visual homogeneity.

Third, with regard to symmetry detection, most theoretical accounts assume that symmetry is explicitly detected using symmetry detectors along particular axes (Wagemans, 1997 [↗](#); Bertamini and Makin, 2014 [↗](#)). By contrast, our findings indicate an indirect mechanism for symmetry detection that does not invoke any special symmetry computations. We show that visual homogeneity computations can easily discriminate between symmetric and asymmetric objects. This is because symmetric objects have high visual homogeneity since they have repeated parts, whereas asymmetric objects have low visual homogeneity since they have disparate parts (Pramod and Arun, 2018 [↗](#)). In a recent study, symmetry detection was explained by the average distance of objects relative to other objects (Pramod and Arun, 2018 [↗](#)). This finding is consistent with ours since visual homogeneity is correlated with the average distance to other objects (Section S1). However, there is an important distinction between these two quantities. Visual homogeneity is an intrinsic image property, whereas the average distance of an object to other objects depends on the set of other objects on which the average is being computed. Indeed, we have confirmed through additional experiments that visual homogeneity is independent of experimental context (Section S4). We speculate that visual homogeneity can explain many other aspects of symmetry perception, such as the relative strength of symmetries.

Visual homogeneity in other visual tasks

Our finding that visual homogeneity explains generic visual tasks has several important implications for visual tasks in general. First, we note that visual homogeneity can be easily extended to explain other generic tasks such as delayed match-to-sample tasks or n-back tasks, by taking the response to the test stimulus as being averaged with the sample-related information in working memory. In such tasks, visual homogeneity will be larger for sequences with repeated compared to non-repeated stimuli, and can easily be used to solve the task. Testing these possibilities will require comparing systematic variations in response times in these tasks across images, and measurements of perceptual space for calculating visual homogeneity.

Second, we note that visual homogeneity can also be extended to explain object categorization, if one assumes that the center in perceptual space for calculating visual homogeneity can be temporarily shifted to the center of an object category. In such tasks, visual homogeneity relative to the category center will be small for objects belonging to a category and large for objects outside the category, and can be used as a decision variable to solve categorization tasks. This idea is consistent with prevalent accounts of object categorization (Stewart and Morin, 2007 [↗](#); Ashby and Maddox, 2011 [↗](#); Mohan and Arun, 2012 [↗](#)). Indeed, categorization response times can be explained using perceptual distances to category and non-category items (Mohan and Arun, 2012 [↗](#)). By reanalyzing data from this study, we have found that, at least for the animate categories tested, visual homogeneity can explain categorization responses (Section S9). However, this remains to be tested in a more general fashion across multiple object categories.

Neural encoding of visual homogeneity

We have found that visual homogeneity is encoded in a specific region of the brain, which we denote as region VH, in both visual search and symmetry detection tasks (Figure 3D [↗](#), 5D). This finding is consistent with observations of norm-based encoding in IT neurons (Leopold et al., 2006) and in face recognition (Valentine, 1991 [↗](#); Rhodes and Jeffery, 2006 [↗](#); Carlin and Kriegeskorte, 2017 [↗](#)). However, our findings are significant because they reveal a dedicated region in high-level visual cortex for solving generic visual tasks.

We have found that the VH region is located just anterior to the lateral occipital (LO) region, where neural dissimilarities match closely with perceptual dissimilarities (**Figure 3E**, **5E**). Based on this proximity, we speculate that visual homogeneity computations are based on object representations in LO. However, confirming this prediction will require fine-grained recordings of neural activity in VH and LO. An interesting possibility for future studies would be to causally perturb brain activity separately in VH or LO using magnetic or electrical stimulation, if at all possible. A simple prediction would be that perturbing LO would distort the underlying representation, whereas perturbing VH would distort the underlying decision process.

We caution however that the results might not be so easily interpretable if visual homogeneity computations in VH are based on object representations in LO.

Recent observations from neural recordings in monkeys suggest that perceptual dissimilarities and visual homogeneity need not be encoded in separate regions. For instance, the overall magnitude of the population neural response of monkey inferior temporal (IT) cortex neurons was found to correlate with memorability (Jaegle et al., 2019). These results are consistent with encoding of visual homogeneity in these regions. However, we note that neural responses in IT cortex also predict perceptual dissimilarities (Op de Beeck et al., 2001; Sripathi and Olson, 2010; Zhivago and Arun, 2014; Agrawal et al., 2020). Taken together, these findings suggest that visual homogeneity computations and the underlying perceptual representation could be interleaved within a single neural population, unlike in humans where we found separate regions. Indeed, in our study, the mean activations of the LO region were also correlated with visual homogeneity for symmetry detection (**Figure 5A**), but not for target present/absent search (**Figure 3A**). We speculate that perhaps visual homogeneity might be intermingled into the object representation in monkeys but separated into a dedicated region in humans. These are interesting possibilities for future work.

Although many previous studies have reported brain activations in the vicinity of the VH region, we are unaware of any study that has ascribed a specific function to this region. The localized activations in our study match closely with the location of the recently reported ventral stream attention module in both humans and monkeys (Sani et al., 2021). Previous studies have observed important differences in brain activations in this region, which can be explained using visual homogeneity, as detailed below.

First, previous studies have observed differences in brain activations for animate compared to inanimate objects in high-level visual areas which have typically included the newly defined VH region reported here (Bracci and Op de Beeck, 2015; Proklova et al., 2016; Thorat et al., 2019). These observations would be consistent with our findings if visual homogeneity is smaller for animate compared to inanimate objects, which would result in weaker activations for animate objects in region VH. Indeed, visual homogeneity, as indexed by the reciprocal of target-absent search time, is smaller for animate objects compared to inanimate objects (Section S9). Likewise, brain activations were weaker for animate objects compared to inanimate objects in region VH (average VH activations, mean $\pm$ sd across participants: 0.50 ± 0.61 for animate target-absent displays, 0.64 ± 0.59 for inanimate target-absent displays, $p < 0.05$, sign-rank test across participants). Based on this we speculate that visual homogeneity may partially explain the animacy organization of human ventral temporal cortex. Establishing this will require controlling animate/inanimate stimuli not only for shape but also for visual homogeneity.

Second, previous studies have reported larger brain activations for symmetric objects compared to asymmetric objects in the vicinity of this region (Sasaki et al., 2005; Van Meel et al., 2019). This can be explained by our finding that symmetric objects have larger visual homogeneity (**Figure 4E**), leading to activation of the VH region (**Figure 5A**). But the increased activations in previous studies were located in the V4 & LO regions, whereas we have found greater activations more anteriorly in the VH region. This difference could be due to the stimulus-related differences:

both previous studies used dot patterns, which could appear more object-like when symmetric, leading to more widespread differences in brain activation due to other visual processes like figure-ground segmentation (Van Meel et al., 2019 [DOI](#)). By contrast, both symmetric and asymmetric objects in our study are equally object-like. Resolving these discrepancies will require measuring visual homogeneity as well as behavioural performance during symmetry detection for dot patterns.

Finally, our results are consistent with the greater activity observed for objects with shared features observed in ventral temporal cortex during naming tasks (Taylor et al., 2012 [DOI](#); Tyler et al., 2013 [DOI](#)). Our study extends these observations by demonstrating an empirical measure for shared feature (target-absent times in visual search) and encoding of this empirical measure into a localized region in object selective cortex across many tasks. We speculate that visual homogeneity may at least partially explain semantic concepts such as those described in these studies.

Relation to image memorability

We have defined a novel image property, visual homogeneity, which refers to the distance of a visual image to a central point in the underlying perceptual representation. It can be reliably estimated for each image as the inverse of the target-absent response time in a visual search task (Figure 2 [DOI](#)) and is independent of the immediate experimental context (Section S4).

The above observations raise the question of whether and how visual homogeneity might be related to image memorability. It has long been noted that faces that are rated as being distinctive or unusual are also easier to remember (Murdock, 1960 [DOI](#); Valentine and Bruce, 1986a [DOI](#), 1986b [DOI](#); Valentine, 1991 [DOI](#)). Recent studies have elucidated this observation by showing that there are specific image properties that predict image memorability (Bainbridge et al., 2017 [DOI](#); Lukavský and Děchtěrenko, 2017 [DOI](#); Rust and Mehrpour, 2020 [DOI](#)). However, image memorability, as elegantly summarized in a recent review (Rust and Mehrpour, 2020 [DOI](#)), could be driven by a number of both intrinsic and extrinsic factors. It would therefore be interesting to characterize how well visual homogeneity, empirically measured using target-absent visual search, can predict image memorability.

Conclusions

Taken together, our results show that a variety of generic visual tasks that require detecting feature-independent image properties, can be solved using visual homogeneity as a decision variable, which is localized to a specific region anterior to the lateral occipital region. While this does not explain all possible variations of these tasks, our study represents an important first step in terms of demonstrating a quantitative, falsifiable model and a localized neural substrate. These can serve as a basis to elucidate whether/how visual homogeneity computations might contribute to a variety of visual tasks.

Materials and methods

All participants had a normal or corrected-to-normal vision and gave informed consent to an experimental protocol approved by the Institutional Human Ethics Committee of the Indian Institute of Science (IHEC # 6-15092017). Participants provided written informed consent before each experiment and were monetarily compensated.

Experiment 1. Oddball detection for perceptual space (natural objects)

Participants

A total of 16 participants (8 males, 22 ± 2.8 years) participated in this experiment.

Stimuli

The stimulus set comprised a set of 32 grayscale natural objects (16 animate, 16 inanimate) presented against a black background.

Procedure

Participants performed an oddball detection task with a block of practice trials involving unrelated stimuli followed by the main task. Each trial began with a red fixation cross (diameter 0.5°) for 500 ms, followed by a 4×4 search array measuring $30^\circ \times 30^\circ$ for 5 seconds or until a response was made. The search array always contained one oddball target and 15 identical distractors, with the target appearing equally often on the left or right. A vertical red line divided the screen equally into two halves to facilitate responses. Participants were asked to respond as quickly and as accurately as possible using a key press to indicate the side of the screen containing the target ('Z' for left, 'M' for right). Incorrect trials were repeated later after a random number of other trials. Each participant completed 992 correct trials ($^{32}C_2$ object pairs $\times$ 2 repetitions with either image as target). The experiment was created using PsychoPy (Peirce et al., 2019 [↗](#)) and ported to the online platform Pavlovia for collecting data.

Since stimulus size could vary with the monitor used by the online participants, we equated the stimulus size across participants using the ScreenScale function (<https://doi.org/10.17605/OSF.IO/8FHQK> [↗](#)). Each participant adjusted the size of a rectangle on the screen such that its size matched the physical dimensions of a credit card. All the stimuli presented were then scaled with the estimated scaling function to obtain the desired size in degrees of visual angle, assuming an average distance to screen of 60 cm.

Data Analysis

Response times faster than 0.3 seconds or slower than 3 seconds were removed from the data. This step removed only 1.25% of the data and improved the overall response time consistency, but did not qualitatively alter the results.

Characterizing perceptual space using multidimensional scaling

To characterize the perceptual space on which present/absent decisions are made, we took the inverse of the average response times (across trials and participants) for each image pair. This inverse of response time (i.e. $1/RT$) represents the dissimilarity between the target and distractor (Arun, 2012 [↗](#)), indexes the underlying salience signal in visual search (Sunder and Arun, 2016 [↗](#)) and combines linearly across a variety of factors (Pramod and Arun, 2014 [↗](#), 2016 [↗](#); Jacob and Arun, 2020 [↗](#)). Since there were 32 natural objects in the experiment and all possible ($^{32}C_2 = 496$) pairwise searches in the experiment, we obtained 496 pairwise dissimilarities overall. To calculate target-present and target-absent array responses, we embedded these objects into a multidimensional space using multidimensional scaling analysis (*mdscale* function; MATLAB 2019). This analysis finds the n -dimensional coordinates for each object such that pairwise distances between objects best matches with the experimentally observed pairwise distances. We

then treated the activations of objects along each dimension as the responses of a single artificial neuron, so that the response to target-present arrays could be computed as the average of the target and distractor responses.

Experiment 2. Target present-absent search during fMRI

Participants

A total of 16 subjects (11 males; age, mean $\pm$ sd: 25 $\pm$ 2.9 years) participated in this experiment. Participants with history of neurological or psychiatric disorders, or with metal implants or claustrophobia were excluded through screening questionnaires.

Procedure

Inside the scanner, participants performed a single run of a one-back task for functional localizers (block design, object vs scrambled objects), eight runs of the present-absent search task (event-related design), and an anatomical scan. The experiment was deployed using custom MATLAB scripts written using Psychophysics Toolbox (Brainard, 1997 [link](#)).

Functional localizer runs

Participants had to view a series of images against a black background and press a response button whenever an item was repeated. On each trial, 16 images were presented (0.8 s on, 0.2 s off), containing one repeat of an image that could occur at random. Trials were combined into blocks of 16 trials each containing either only objects or only scrambled objects. A single run of the functional localizers contained 12 such blocks (6 object blocks & 6 scrambled-object blocks). Stimuli in each block were chosen randomly from a larger pool of 80 naturalistic objects with the corresponding phase-scrambled objects (created by taking the 2D Fourier transform of each image, randomly shuffling the Fourier phase, and performing the Fourier inverse transform). This is a widely used method for functional localization of object-selective cortex. In practice, however, we observed no qualitative differences in our results upon using voxels activated during these functional localizer runs to further narrow down the voxels selected using anatomical masks. As a result, we did not use the functional localizer data, and all the analyses presented here are based on anatomical masks only.

Visual search task

In the present-absent search task, participants reported the presence or absence of an oddball target by pressing one of two buttons using their right hand. The response buttons were fixed for a given participant and counterbalanced across participants. Each search array had eight items, measuring 1.5° along the longer dimension, arranged in a 3 x 3 grid, with no central element to avoid fixation biases (as shown in **Figure 2C** [link](#)). The entire search array measured 6.5°, with an average inter-item spacing of 2.5°. Item positions were jittered randomly on each trial according to a uniform distribution with range $\pm$ 0.2°. Each trial lasted 4 s (1 s ON time and 3 s OFF time), and participants had to respond within 4 s. Each run had 64 unique searches (32 present, 32 absent) presented in random order, using the natural objects from Experiment 1. Target-present searches were chosen randomly from all possible searches such that all 32 images appeared equally often. Target-absent searches included all 32 objects. The location of the target in the target-present searches was chosen such that all eight locations were sampled equally often. In this manner, participants performed 8 such runs of 64 trials each.

Data acquisition

Participants viewed images projected on a screen through a mirror placed above their eyes. Functional MRI (fMRI) data were acquired using a 32-channel head coil on a 3T Skyra (Siemens, Mumbai, India) at the HealthCare Global Hospital, Bengaluru. Functional scans were performed using a T2*-weighted gradient-echo-planar imaging sequence with the following parameters: repetition time (TR) = 2s, echo time (TE) = 28 ms, flip angle = 79°, voxel size = $3 \times 3 \times 3 \text{ mm}^3$, field of view = $192 \times 192 \text{ mm}^2$, and 33 axial-oblique slices for whole-brain coverage. Anatomical scans were performed using T1-weighted images with the following parameters: TR = 2.30 s, TE = 1.99 ms, flip angle = 9°, voxel size = $1 \times 1 \times 1 \text{ mm}^3$, field of view = $256 \times 256 \times 176 \text{ mm}^3$.

Data preprocessing

The raw fMRI data were preprocessed using Statistical Parametric Mapping (SPM) software (Version12; Wellcome Center for Human Neuroimaging; <https://www.fil.ion.ucl.ac.uk/spm/software/spm12/>), running on MATLAB 2019b. Raw images were realigned, slice-time corrected, co-registered to the anatomical image, segmented, and normalized to the Montreal Neurological Institute (MNI) 305 anatomical template. Repeating the key analyses with voxel activations estimated from individual subjects yielded qualitatively similar results. Smoothing was performed only on the functional localizer blocks using a Gaussian kernel with a full-width half-maximum of 5 mm. Default SPM parameters were used, and voxel size after normalization was kept at $3 \times 3 \times 3 \text{ mm}^3$. The data were further processed using GLMdenoise (Kay et al., 2013). GLMdenoise improves the signal-to-noise ratio in the data by regressing out the noise estimated from task-unrelated voxels. The denoised time-series data were modeled using generalized linear modeling in SPM after removing low-frequency drift using a high-pass filter with a cutoff of 128 s. In the main experiment, the activity of each voxel was modeled using 83 regressors (68 stimuli + 1 fixation + 6 motion regressors + 8 runs). In the localizer block, each voxel was modeled using 14 regressors (6 stimuli + 1 fixation + 6 motion regressors + 1 run).

ROI definitions

The regions of interest (ROI) of Early Visual Cortex (EVC) and Lateral Occipital (LO) regions were defined using anatomical masks from the SPM anatomy toolbox (Eickhoff et al., 2005). All brain maps were visualized on the inflated brain using Freesurfer (<https://surfer.nmr.mgh.harvard.edu/fswiki/>).

Behavioral data analysis

Response times faster than 0.3 seconds or slower than 3 seconds were removed from the data. This step removed only 0.75% of the data and improved the overall response time consistency, but did not qualitatively alter the results.

Model fitting for visual homogeneity

We took the multi-dimensional embedding returned by the perceptual space experiment (Experiment 1) in 5 dimensions as the responses of 5 artificial neurons to the entire set of objects. For each target-present array, we calculated the neural response as the average of the responses elicited by these 5 neurons to the target and distractor items. Likewise, for target-absent search arrays, the neural response was simply the response elicited by these 5 neurons to the distractor item in the search array. To estimate the visual homogeneity of the target-present and target-absent search arrays, we calculated the distance of each of these arrays from a single point in the multidimensional representation. We then calculated the correlation between the visual homogeneity calculated relative to this point and the response times for the target-present and target-absent search arrays. The 5 coordinates of this center point was adjusted using constrained nonlinear optimization to maximize the difference between correlations with the target-present &

target-absent response times respectively. This optimum center remained stable across many random starting points, and our results were qualitatively similar upon varying the number of embedding dimensions.

Additionally, we performed a leave-one-out cross-validation analysis to validate the number of dimensions or neurons used for the multidimensional scaling analysis in the visual homogeneity model fits. For each choice of number of dimensions, we estimated the optimal centre for visual homogeneity calculations while leaving out all searches involving a single image. We then calculated the visual homogeneity for all the target-present and target-absent searches involving the left-out image. Compiling these predictions by leaving out all images by turn results in a leave-one-out predicted visual homogeneity, which we correlated with the target-present and target-absent response times. We found that the absolute sum of the correlations between visual homogeneity and present/absent reaction times increased monotonically from 1 to 5 neurons, remained at a steady level from 5 to 9 neurons and decreased beyond 9 neurons. Furthermore, the visual homogeneity using the optimal center is highly correlated for 5-9 neurons. We therefore selected 5 neurons or dimensions for reporting visual homogeneity computations.

Searchlight maps for mean activation (Figure 3A, Figure 5A)

To characterize the brain regions that encode visual homogeneity, we performed a whole-brain searchlight analysis. For each voxel, we took the voxels in a 3x3x3 neighborhood and calculated the mean activations across these voxels across all participants. To avoid overall activation level differences between target-present and target-absent searches, we z-scored the mean activations separately across target-present and target-absent searches. Similarly, we calculated the visual homogeneity model predictions from behaviour, and z-scored the visual homogeneity values for target-present and target-absent searches separately. We then calculated the correlation between the normalized mean activations and the normalized visual homogeneity for each voxel, and displayed this as a colormap on the inflated MNI brain template in **Figures 3A** & **5A**.

Note that the z-scoring of mean activations and visual homogeneity removes any artefactual correlation between mean activation and visual homogeneity arising simply due to overall level differences in mean activation or visual homogeneity itself, but does not alter the overall positive correlation between the visual homogeneity and mean activation across individual search displays.

Searchlight maps for neural and behavioural dissimilarity (Figure 3B, Figure 5B)

To characterize the brain regions that encode perceptual dissimilarity, we performed a whole-brain searchlight analysis. For each voxel, we took the voxel activations in a 3x3x3 neighborhood to target-absent displays as a proxy for the neural response to the single image. For each image pair, we calculated the pair-wise Euclidean distance between the 27-dimensional voxel activations evoked by the two images, and averaged this distance across participants to get a single average distance. For 32 target-absent displays in the experiment, taking all possible pairwise distances results in $^{32}C_2 = 496$ pairwise distances. Similarly, we obtained the same 496 pairwise perceptual dissimilarities between these items from the oddball detection task (Experiment 1). We then calculated the correlation between the mean neural dissimilarities at each voxel with perceptual dissimilarities, and displayed this as a colormap on the flattened MNI brain template in **Figures 3B** & **5B**.

Experiment 3. Oddball detection for perceptual space (Symmetric/Asymmetric objects)

Participants

A total of 15 participants (11 males, 22.8 ± 4.3 years) participated in this experiment.

Paradigm

Participants performed an oddball visual search task. Participants completed 4032 correct trials ($^{64}C_2$ shape pairs x 2 repetitions) as two sessions in two days. We used a total of 64 baton shapes (32 symmetric and 32 asymmetric), and all shapes were presented against a black background. We created 32 unique parts with the vertical line as part of the contour. We created 32 symmetric by joining the part and its mirror-filled version, and 32 asymmetric objects were created by randomly pairing the left part and mirror flipped version of another left part. All parts were occurring equally likely. All other task details are the same as Experiment 1.

Experiment 4 Symmetry judgment task (fMRI & behavior)

Participants

A total of 15 subjects participated in this study. Participants had normal or corrected to normal vision. Participants had no history of neurological or psychiatric impairment. We excluded participants with metal implants or claustrophobia from the study.

Paradigm

Inside the scanner, participants performed two runs of one-back identity task (functional localizer), eight runs of symmetry judgment task (event-related design), and one anatomical scan. We excluded the data from one participant due to poor accuracy and long response times.

Symmetry Task

On each trial, participants had to report whether a briefly presented object was symmetric or not using a keypress. Objects measured 4° and were presented against a black background. Response keys were counterbalanced across participants. Each trial lasted 4 s, with the object displayed for 200 ms followed by a blank period of 3800 ms. Participants could respond at any time following appearance of the object, up to a time out of 4 s after which the next trial began. Each run had 64 unique conditions (32 symmetric and 32 asymmetric).

1-back task for functional localizers

Stimuli were presented as blocks, and participants reported repeated stimuli with a keypress. Each run had blocks of either silhouettes (asymmetric/symmetric), dot patterns (asymmetric/symmetric), combination of baton and dot patterns (asymmetric/symmetric) and natural objects (intact/scrambled).

Data Analysis

Noise Removal in RT

Very fast (< 100 ms) reaction times were removed. We also discarded all reaction times to an object if participant's accuracy was less than 80%. This process removed 3.6% of RT data.

Model fitting for visual homogeneity

We proceeded as before by embedding the oddball detection response times into multidimensional space with 3 dimensions. For each image, the visual homogeneity was defined as its distance from an optimum center. We then calculated the correlation between the visual homogeneity calculated relative to this optimum center and the response times for the target-present and target-absent search arrays separately. This optimum center was estimated using a constrained nonlinear optimization to maximize the difference between the correlations for asymmetric object response times and symmetric object response times. Other details were the same as in Experiment-2.

Data availability

All data and code required to reproduce the results will be made available on OSF.

Author contributions

GJ and SPA designed the visual search experiments; GJ collected behavioral and fMRI data; GJ & SPA analyzed and interpreted the data; PRT & SPA designed the symmetry experiments; PRT collected fMRI data; GJ, PRT & SPA analyzed and interpreted the data; GJ and SPA wrote the manuscript with inputs from PRT.

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Article and author information

Georgin Jacob

Department of Electrical Communication Engineering Indian Institute of Science, Bangalore 560012, Centre for Neuroscience Indian Institute of Science, Bangalore 560012
ORCID iD: [0000-0001-8262-0155](https://orcid.org/0000-0001-8262-0155)

R. T. Pramod

Department of Electrical Communication Engineering Indian Institute of Science, Bangalore 560012, Centre for Neuroscience Indian Institute of Science, Bangalore 560012
ORCID iD: [0000-0002-5933-7893](https://orcid.org/0000-0002-5933-7893)

S. P. Arun

Centre for Neuroscience Indian Institute of Science, Bangalore 560012, Department of Electrical Communication Engineering Indian Institute of Science, Bangalore 560012
For correspondence: sparun@iisc.ac.in
ORCID iD: [0000-0001-9602-5066](https://orcid.org/0000-0001-9602-5066)

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Reviewer #1 (Public Review):

Summary:

The authors define a new metric for visual displays, derived from psychophysical response times, called visual homogeneity (VH). They attempt to show that VH is explanatory of response times across multiple visual tasks. They use fMRI to find visual cortex regions with VH-correlated activity. On this basis, they declare a new visual region in human brain, area VH, whose purpose is to represent VH for the purpose of visual search and symmetry tasks.

Strengths:

The authors present carefully designed experiments, combining multiple types of visual judgments and multiple types of visual stimuli with concurrent fMRI measurements. This is a rich dataset with many possibilities for analysis and interpretation.

Weaknesses:

The datasets presented here should provide a rich basis for analysis. However, in this version of the manuscript, I believe that there are major problems with the logic underlying the authors' new theory of visual homogeneity (VH), with the specific methods they used to calculate VH, and with their interpretation of psychophysical results using these methods. These problems with the coherency of VH as a theoretical construct and metric value make it hard to interpret the fMRI results based on searchlight analysis of neural activity correlated with VH. In addition, the large regions of VH correlations identified in Experiments 1 and 2 vs. Experiments 3 and 4 are barely overlapping. This undermines the claim that VH is a universal quantity, represented in a newly discovered area of visual cortex, that underlies a wide variety of visual tasks and functions.

Maybe I have missed something, or there is some flaw in my logic. But, absent that, I think the authors should radically reconsider their theory, analyses, and interpretations, in light of detailed comments below, in order to make the best use of their extensive and valuable datasets combining behavior and fMRI. I think doing so could lead to a much more coherent and convincing paper, albeit possibly supporting less novel conclusions.

THEORY AND ANALYSIS OF VH

(1) VH is an unnecessary, complex proxy for response time and target-distractor similarity.

VH is defined as a novel visual quality, calculable for both arrays of objects (as studied in Experiments 1-3) and individual objects (as studied in Experiment 4). It is derived from a center-to-distance calculation in a perceptual space. That space in turn is derived from multi-dimensional scaling of response times for target-distractor pairs in an oddball detection task (Experiments 1 and 2) or in a same different task (Experiments 3 and 4). Proximity of objects in the space is inversely proportional to response times for arrays in which they were paired.

These response times are higher for more similar objects. Hence, proximity is proportional to similarity. This is visible in Fig. 2B as the close clustering of complex, confusable animal shapes.

VH, i.e. distance-to-center, for target-present arrays is calculated as shown in Fig. 1C, based on a point on the line connecting target and distractors. The authors justify this idea with previous findings that responses to multiple stimuli are an average of responses to the constituent individual stimuli. The distance of the connecting line to the center is inversely proportional to the distance between the two stimuli in the pair, as shown in Fig. 2D. As a result, VH is inversely proportional to distance between the stimuli and thus to stimulus similarity and response times. But this just makes VH a highly derived, unnecessarily complex proxy for target-distractor similarity and response time. The original response times on which the perceptual space is based are far more simple and direct measures of similarity for predicting response times.

(2) The use of VH derived from Experiment 1 to predict response times in Experiment 2 is circular and does not validate the VH theory.

The use of VH, a response time proxy, to predict response times in other, similar tasks, using the same stimuli, is circular. In effect, response times are being used to predict response times across two similar experiments using the same stimuli. Experiment 1 and the target present condition of Experiment 2 involve the same essential task of oddball detection. The results of Experiment 1 are converted into VH values as described above, and these are used to predict response times in experiment 2 (Fig. 2F). Since VH is a derived proxy for response values in Experiment 1, this prediction is circular, and the observed correlation shows only consistency between two oddball detection tasks in two experiments using the same stimuli.

(3) The negative correlation of target-absent response times with VH as it is defined for target-absent arrays, based on distance of a single stimulus from center, is uninterpretable without understanding the effects of center-fitting. Most likely, center-fitting and the different VH metric for target-absent trials produce an inverse correlation of VH with target-distractor similarity.

The construction of the VH perceptual space also involves fitting a "center" point such that distances to center predict response times as closely as possible. The effect of this fitting process on distance-to-center values for individual objects or clusters of objects is unknowable from what is presented here. These effects would depend on the residual errors after fitting response times with the connecting line distances. The center point location and its effects on distance-to-center of single objects and object clusters are not discussed or reported here.

Yet, this uninterpretable distance-to-center of single objects is chosen as the metric for VH of target-absent displays (VHabsent). This is justified by the idea that arrays of a single stimulus will produce an average response equal to one stimulus of the same kind. But it is not logically clear why response strength to a stimulus should be a metric for homogeneity of arrays constructed from that stimulus, or even what homogeneity could mean for a single stimulus from this set. And it is not clear how this VHabsent metric based on single stimuli can be equated to the connecting line VH metric for stimulus pairs, i.e. VHpresent, or how both could be plotted on a single continuum.

It is clear, however, what *should* be correlated with difficulty and response time in the target-absent trials, and that is the complexity of the stimuli and the numerosity of similar distractors in the overall stimulus set. Complexity of the target, similarity with potential distractors, and number of such similar distractors all make ruling out distractor presence more difficult. The correlation seen in Fig. 2G must reflect these kinds of effects, with higher

response times for complex animal shapes with lots of similar distractors and lower response times for simpler round shapes with fewer similar distractors.

The example points in Fig. 2G seem to bear this out, with higher response times for the deer stimulus (complex, many close distractors in the Fig. 2B perceptual space) and lower response times for the coffee cup (simple, few close distractors in the perceptual space). While the meaning of the VH scale in Fig. 2G, and its relationship to the scale in Fig. 2F, are unknown, it seems like the Fig. 2G scale has an inverse relationship to stimulus complexity, in contrast to the expected positive relationship for Fig. 2F. This is presumably what creates the observed negative correlation in Fig. 2G.

Taken together, points 1-3 suggest that VH_{present} and VH_{absent} are complex, unnecessary, and disconnected metrics for understanding target detection response times. The standard, simple explanation should stand. Task difficulty and response time in target detection tasks, in both present and absent trials, are positively correlated with target-distractor similarity.

I think my interpretations apply to Experiments 3 and 4 as well, although I find the analysis in Fig. 4 especially hard to understand. The VH space in this case is based on Experiment 3 oddball detection in a stimulus set that included both symmetric and asymmetric objects. But the response times for a very different task in Experiment 4, a symmetric/asymmetric judgment, are plotted against the axes derived from Experiment 3 (Fig. 4F and 4G). It is not clear to me why a measure based on oddball detection that requires no use of symmetry information should be predictive of within-stimulus symmetry detection response times. If it is, that requires a theoretical explanation not provided here.

(4) Contrary to the VH theory, same/different tasks are unlikely to depend on a decision boundary in the middle of a similarity or homogeneity continuum.

The authors interpret the inverse relationship of response times with VH_{present} and VH_{absent} , described above, as evidence for their theory. They hypothesize, in Fig. 1G, that VH_{present} and VH_{absent} occupy a single scale, with maximum VH_{present} falling at the same point as minimum VH_{absent} . This is not borne out by their analysis, since the VH_{present} and VH_{absent} value scales are mainly overlapping, not only in Experiments 1 and 2 but also in Experiments 3 and 4. The authors dismiss this problem by saying that their analyses are a first pass that will require future refinement. Instead, the failure to conform to this basic part of the theory should be a red flag calling for revision of the theory.

The reason for this single scale is that the authors think of target detection as a boundary decision task, along a single scale, with a decision boundary somewhere in the middle, separating present and absent. This model makes sense for decision dimensions or spaces where there are two categories (right/left motion; cats vs. dogs), separated by an inherent boundary (equal left/right motion; training-defined cat/dog boundary). In these cases, there is less information near the boundary, leading to reduced speed/accuracy and producing a pattern like that shown in Fig. 1G.

This logic does not hold for target detection tasks. There is no inherent middle point boundary between target present and target absent. Instead, in both types of trial, maximum information is present when target and distractors are most dissimilar, and minimum information is present when target and distractors are most similar. The point of greatest similarity occurs at the limit of any metric for similarity. Correspondingly, there is no middle point dip in information that would produce greater difficulty and higher response times. Instead, task difficulty and response times increase monotonically with similarity between targets and distractors, for both target present and target absent decisions. Thus, in Figs. 2F and 2G, response times appear to be highest for animals, which share the largest numbers of closely similar distractors.

DEFINITION OF AREA VH USING fMRI

(1) The area VH boundaries from different experiments are nearly completely non-overlapping.

In line with their theory that VH is a single continuum with a decision boundary somewhere in the middle, the authors use fMRI searchlight to find an area whose responses positively correlate with homogeneity, as calculated across all of their target present and target absent arrays. They report VH-correlated activity in regions anterior to LO. However, the VH defined by symmetry Experiments 3 and 4 (VHsymmetry) is substantially anterior to LO, while the VH defined by target detection Experiments 1 and 2 (VHdetection) is almost immediately adjacent to LO. Fig. S13 shows that VHsymmetry and VHdetection are nearly non-overlapping. This is a fundamental problem with the claim of discovering a new area that represents a new quantity that explains response times across multiple visual tasks. In addition, it is hard to understand why VHsymmetry does not show up in a straightforward subtraction between symmetric and asymmetric objects, which should show a clear difference in homogeneity.

(2) It is hard to understand how neural responses can be correlated with both VHpresent and VHabsent.

The main paper results for VHdetection are based on both target-present and target-absent trials, considered together. It is hard to interpret the observed correlations, since the VHpresent and VHabsent metrics are calculated in such different ways and have opposite correlations with target similarity, task difficulty, and response times (see above). It may be that one or the other dominates the observed correlations. It would be clarifying to analyze correlations for target-present and target-absent trials separately, to see if they are both positive and correlated with each other.

(3) Definition of the boundaries and purpose of a new visual area in the brain requires circumspection, abundant and convergent evidence, and careful controls.

Even if the VH metric, as defined and calculated by the authors here, is a meaningful quantity, it is a bold claim that a large cortical area just anterior to LO is devoted to calculating this metric as its major task. Vision involves much more than target detection and symmetry detection. Cortex anterior to LO is bound to perform a much wider range of visual functionalities. If the reported correlations can be clarified and supported, it would be more circumspect to treat them as one byproduct of unknown visual processing in cortex anterior to LO, rather than treating them as the defining purpose for a large area of visual cortex.

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Reviewer #3 (Public Review):

Summary:

This study proposes visual homogeneity as a novel visual property that enables observers perform to several seemingly disparate visual tasks, such as finding an odd item, deciding if two items are same, or judging if an object is symmetric. In Exp 1, the reaction times on several objects were measured in human subjects. In Exp 2, visual homogeneity of each object was calculated based on the reaction time data. The visual homogeneity scores predicted reaction times. This value was also correlated with the BOLD signals in a specific region anterior to LO. Similar methods were used to analyze reaction time and fMRI data in a symmetry detection task. It is concluded that visual homogeneity is an important feature that enables observers to solve these two tasks.

Strengths:

- (1) The writing is very clear. The presentation of the study is informative.
- (2) This study includes several behavioral and fMRI experiments. I appreciate the scientific rigor of the authors.

Weaknesses:

(1) My main concern with this paper is the way visual homogeneity is computed. On page 10, lines 188-192, it says: "we then asked if there is any point in this multidimensional representation such that distances from this point to the target-present and target-absent response vectors can accurately predict the target-present and target-absent response times with a positive and negative correlation respectively (see Methods)". This is also true for the symmetry detection task. If I understand correctly, the reference point in this perceptual space was found by deliberating satisfying the negative and positive correlations in response times. And then on page 10, lines 200-205, it shows that the positive and negative correlations actually exist. This logic is confusing. The positive and negative correlations emerge only because this method is optimized to do so. It seems more reasonable to identify the reference point of this perceptual space independently, without using the reaction time data. Otherwise, the inference process sounds circular. A simple way is to just use the mean point of all objects in Exp 1, without any optimization towards reaction time data.

(2) Visual homogeneity (at least given the current from) is an unnecessary term. It is similar to distractor heterogeneity/distractor variability/distractor statics in literature. However, the authors attempt to claim it as a novel concept. The title is "visual homogeneity computations in the brain enable solving generic visual tasks". The last sentence of the abstract is "a NOVEL IMAGE PROPERTY, visual homogeneity, is encoded in a localized brain region, to solve generic visual tasks". In the significance, it is mentioned that "we show that these tasks can be solved using a simple property WE DEFINE as visual homogeneity". If the authors agree that visual homogeneity is not new, I suggest a complete rewrite of the title, abstract, significance, and introduction.

(3) Also, "solving generic tasks" is another overstatement. The oddball search tasks, same-different tasks, and symmetric tasks are only a small subset of many visual tasks. Can this "quantitative model" solve motion direction judgment tasks, visual working memory tasks? Perhaps so, but at least this manuscript provides no such evidence. On line 291, it says "we have proposed that visual homogeneity can be used to solve any task that requires discriminating between homogeneous and heterogeneous displays". I think this is a good statement. A title that says "XXXX enable solving discrimination tasks with multi-component displays" is more acceptable. The phrase "generic tasks" is certainly an exaggeration.

(4) If I understand it correctly, one of the key findings of this paper is "the response times for target-present searches were positively correlated with visual homogeneity. By contrast, the response times for target-absent searches were negatively correlated with visual homogeneity" (lines 204-207). I think the authors have already acknowledged that the positive correlation is not surprising at all because it reflects the classic target-distractor similarity effect. But the authors claim that the negative correlations in target-absent searches is the true novel finding.

(5) I would like to make it clear that this negative correlation is not new either. The seminal paper by Duncan and Humphreys (1989) has clearly stated that "difficulty increases with increased similarity of targets to nontargets and decreased similarity between nontargets" (the sentence in their abstract). Here, "similarity between nontargets" is the same as the visual homogeneity defined here. Similar effects have been shown in Duncan (1989) and Nagy, Neriani, and Young (2005). See also the inconsistent results in Nagy & Thomas, 2003, Vicent, Baddeley, Troscianko & Gilchrist, 2009. More recently, Wei Ji Ma has systematically investigated the effects of heterogeneous

distractors in visual search. I think the introduction part of Wei Ji Ma's paper (2020) provides a nice summary of this line of research.

I am surprised that these references are not mentioned at all in this manuscript (except Duncan and Humphreys, 1989).

(6) If the key contribution is the quantitative model, the study should be organized in a different way. Although the findings of positive and negative correlations are not novel, it is still good to propose new models to explain classic phenomena. I would like to mention the three studies by Wei Ji Ma (see below). In these studies, Bayesian observer models were established to account for trial-by-trial behavioral responses. These computational models can also account for the set-size effect, behavior in both localization and detection tasks. I see much more scientific rigor in their studies. Going back to the quantitative model in this paper, I am wondering whether the model can provide any qualitative prediction beyond the positive and negative correlations? Can the model make qualitative predictions that differ from those of Wei Ji's model? If not, can the authors show that the model can quantitatively better account for the data than existing Bayesian models? We should evaluate a model either qualitatively or quantitatively.

(7) In my opinion, one of the advantages of this study is the fMRI dataset, which is valuable because previous studies did not collect fMRI data. The key contribution may be the novel brain region associated with display heterogeneity. If this is the case, I would suggest using a more parametric way to measure this region. For example, one can use Gabor stimuli and systematically manipulate the variations of multiple Gabor stimuli, the same logic also applies to motion direction. If this study uses static Gabor, random dot motion, object images that span from low-level to high-level visual stimuli, and consistently shows that the stimulus heterogeneity is encoded in one brain region, I would say this finding is valuable. But this sounds like another experiment. In other words, it is insufficient to claim a new brain region given the current form of the manuscript.

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Author response:

The following is the authors' response to the original reviews.

eLife assessment:

This study uses carefully designed experiments to generate a useful behavioural and neuroimaging dataset on visual cognition. The results provide solid evidence for the involvement of higher-order visual cortex in processing visual oddballs and asymmetry. However, the evidence provided for the very strong claims of homogeneity as a novel concept in vision science, separable from existing concepts such as target saliency, is inadequate.

We appreciate the positive and balanced assessment from the reviewers. We agree that visual homogeneity is similar to existing concepts such as target saliency. We have tried our best to articulate our rationale for defining it as a novel concept. However, the debate about whether visual homogeneity is novel or related to existing concepts is completely beside the point, since that is not the key contribution of our study.

Our key contribution is our quantitative model for how the brain could be solving generic visual tasks by operating on a feature space. In the literature there are no theories regarding the decision-making process by which the brain could be solving generic visual tasks. In fact, oddball search tasks, same-different tasks and symmetry tasks are never even mentioned in the same study because it is tacitly assumed that the underlying processes are completely different! Our work brings together these disparate tasks by proposing a specific computation that enables the brain to solve both types of tasks and providing evidence for it. This specific computation is a well-defined, falsifiable model that will need to be replicated, elaborated and refined by future studies.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

The authors define a new metric for visual displays, derived from psychophysical response times, called visual homogeneity (VH). They attempt to show that VH is explanatory of response times across multiple visual tasks. They use fMRI to find visual cortex regions with VH-correlated activity. On this basis, they declare a new visual region in the human brain, area VH, whose purpose is to represent VH for the purpose of visual search and symmetry tasks.

Thank you for your concise summary. We appreciate your careful reading and thoughtful and constructive comments.

Strengths:

The authors present carefully designed experiments, combining multiple types of visual judgments and multiple types of visual stimuli with concurrent fMRI measurements. This is a rich dataset with many possibilities for analysis and interpretation.

Thank you for your accurate assessment of the strengths of our study.

Weaknesses:

The datasets presented here should provide a rich basis for analysis. However, in this version of the manuscript, I believe that there are major problems with the logic underlying the authors' new theory of visual homogeneity (VH), with the specific methods they used to calculate VH, and with their interpretation of psychophysical results using these methods. These problems with the coherency of VH as a theoretical construct and

metric value make it hard to interpret the fMRI results based on searchlight analysis of neural activity correlated with VH.

We appreciate your concerns, and have tried our best to respond to them fully against your specific concerns below.

In addition, the large regions of VH correlations identified in Experiments 1 and 2 vs. Experiments 3 and 4 are barely overlapping. This undermines the claim that VH is a universal quantity, represented in a newly discovered area of the visual cortex, that underlies a wide variety of visual tasks and functions.

We agree with you that the VH regions defined using symmetry task and search task do not overlap completely (as we have shown in Figure S13). However this is to be expected for several reasons. First, the images in the symmetry task were presented at fixation, whereas the images in the visual search task were presented peripherally. Second, the lack of overlap could be due to variations across individuals. Indeed, considerable individual variability has been observed in the location of category-selective regions such as VWFA (Glezer and Riesenhuber 2013) and FFA (Weiner and Grill-Spector, 2012). We propose that testing the same participants on both search and symmetry tasks would reveal overlapping VH regions. We now acknowledge these issues in the Results (p. 26).

Maybe I have missed something, or there is some flaw in my logic. But, absent that, I think the authors should radically reconsider their theory, analyses, and interpretations, in light of the detailed comments below, to make the best use of their extensive and valuable datasets combining behavior and fMRI. I think doing so could lead to a much more coherent and convincing paper, albeit possibly supporting less novel conclusions.

We appreciate your concerns. We have tried our best to respond to them fully against your specific concerns below.

THEORY AND ANALYSIS OF VH

(1) VH is an unnecessary, complex proxy for response time and target-distractor similarity. VH is defined as a novel visual quality, calculable for both arrays of objects (as studied in Experiments 1-3) and individual objects (as studied in Experiment 4). It is derived from a center-to-distance calculation in a perceptual space. That space in turn is derived from the multi-dimensional scaling of response times for target-distractor pairs in an oddball detection task (Experiments 1 and 2) or in a same-different task (Experiments 3 and 4).

The above statements are not entirely correct. Experiments 1 & 3 are oddball visual search experiments. Their purpose was to estimate the underlying perceptual space of objects.

Proximity of objects in the space is inversely proportional to response times for arrays in which they were paired. These response times are higher for more similar objects. Hence, proximity is proportional to similarity. This is visible in Fig. 2B as the close clustering of complex, confusable animal shapes.

VH, i.e. distance-to-center, for target-present arrays, is calculated as shown in Fig. 1C, based on a point on the line connecting the target and distractors. The authors justify this idea with previous findings that responses to multiple stimuli are an average of responses to the constituent individual stimuli. The distance of the connecting line to the center is inversely proportional to the distance between the two stimuli in the pair, as shown in Fig. 2D. As a result, VH is inversely proportional to the distance between the stimuli and thus to stimulus similarity and response times. But this just makes VH a

highly derived, unnecessarily complex proxy for target-distractor similarity and response time. The original response times on which the perceptual space is based are far more simple and direct measures of similarity for predicting response times.

We agree that VH brings no explanatory power to target-present searches, since target-present response times are a direct estimate of target-distractor similarity. However, we are additionally explaining target-absent response times. Target-absent response times are well known to vary systematically with image properties, but why they do so have not been clear in the literature.

Our key conceptual advance lies in relating the neural response to a search array to the neural response of the constituent elements, and in proposing a decision variable using which participants can make both target-present and target-absent judgements on any search array.

(2) The use of VH derived from Experiment 1 to predict response times in Experiment 2 is circular and does not validate the VH theory.

The use of VH, a response time proxy, to predict response times in other, similar tasks, using the same stimuli, is circular. In effect, response times are being used to predict response times across two similar experiments using the same stimuli. Experiment 1 and the target present condition of Experiment 2 involve the same essential task of oddball detection. The results of Experiment 1 are converted into VH values as described above, and these are used to predict response times in Experiment 2 (Fig. 2F). Since VH is a derived proxy for response values in Experiment 1, this prediction is circular, and the observed correlation shows only consistency between two oddball detection tasks in two experiments using the same stimuli.

We agree that it would be circular to use oddball search times in Experiment 1 to explain only target-present search times in Experiment 2, since they basically involve the same searches. However, we are explaining both target-present and target-absent search times in a unified framework; systematic variations in target-absent search times have been noted in the literature but never really explained. One could still simply say that target-absent search times are some function of the target-present search times, but this still doesn't provide an explanation for how participants are making target-present and absent decisions. The existing literature contains models for how visual search might occur for a specific target and distractor but does not elucidate how participants might perform generic visual search where target and distractors are not known in advance.

Our key conceptual advance lies in relating the neural response to a search array to the neural response of the constituent elements, and in proposing a decision variable using which participants can make both target-present and target-absent judgements on any search array.

(3) The negative correlation of target-absent response times with VH as it is defined for target-absent arrays, based on the distance of a single stimulus from the center, is uninterpretable without understanding the effects of center-fitting. Most likely, center-fitting and the different VH metrics for target-absent trials produce an inverse correlation of VH with target-distractor similarity.

We see no cause for concern with the center-fitting procedure, for several reasons. First, the best-fitting center remained stable despite many randomly initialized starting points. Second, the best-fitting center derived from one set of objects was able to predict the target-absent and target-present responses of another set of objects. Finally, the VH obtained for each object (i.e. distance from the best-fitting center) is strongly correlated with the average

distance of that object from all other objects (Figure S1A). We have now clarified this in the Results (p. 11).

The construction of the VH perceptual space also involves fitting a "center" point such that distances to center predict response times as closely as possible. The effect of this fitting process on distance-to-center values for individual objects or clusters of objects is unknowable from what is presented here. These effects would depend on the residual errors after fitting response times with the connecting line distances. The center point location and its effects on the distance-to-center of single objects and object clusters are not discussed or reported here.

While it is true that the optimal center needs to be found by fitting to the data, there is no particular mystery to the algorithm: we are simply performing a standard gradient-descent to maximize the fit to the data. We have described the algorithm clearly and are making our codes public. We find the algorithm to yield stable optimal centers despite many randomly initialized starting points. We find the optimal center to be able to predict responses to entirely novel images that were excluded during model training. We are making no assumption about the location of center with respect to individual points. Therefore, we see no cause for concern regarding the center-finding algorithm.

Yet, this uninterpretable distance-to-center of single objects is chosen as the metric for VH of target-absent displays (VHabsent). This is justified by the idea that arrays of a single stimulus will produce an average response equal to one stimulus of the same kind. However, it is not logically clear why response strength to a stimulus should be a metric for homogeneity of arrays constructed from that stimulus, or even what homogeneity could mean for a single stimulus from this set. It is not clear how this VHabsent metric based on single stimuli can be equated to the connecting line VH metric for stimulus pairs, i.e. VHpresent, or how both could be plotted on a single continuum.

Most visual tasks, such as finding an animal, are thought to involve building a decision boundary on some underlying neural representation. Even visual search has been portrayed as a signal-detection problem where a particular target is to be discriminated from a distractor. However none of these formulations work in the case of generic visual tasks, where the target and distractor identities are unknown. We are proposing that, when we view a search array, the neural response to the search array can be deduced from the neural responses to the individual elements using well known rules, and that decisions about an oddball target being present or absent can be made by computing the distance of this neural response from some canonical mean firing rate of a population of neurons. This distance to center computation is what we denote as visual homogeneity. We have revised our manuscript throughout to make this clearer and we hope that this helps you understand the logic better.

It is clear, however, what should be correlated with difficulty and response time in the target-absent trials, and that is the complexity of the stimuli and the numerosity of similar distractors in the overall stimulus set. The complexity of the target, similarity with potential distractors, and the number of such similar distractors all make ruling out distractor presence more difficult. The correlation seen in Fig. 2G must reflect these kinds of effects, with higher response times for complex animal shapes with lots of similar distractors and lower response times for simpler round shapes with fewer similar distractors.

You are absolutely correct that the stimulus complexity should matter, but there are no good measures for stimulus complexity. But considering what factors are correlated with target-

absent response times is entirely different from asking what decision variable or template is being used by participants to solve the task.

The example points in Fig. 2G seem to bear this out, with higher response times for the deer stimulus (complex, many close distractors in the Fig. 2B perceptual space) and lower response times for the coffee cup (simple, few close distractors in the perceptual space). While the meaning of the VH scale in Fig. 2G, and its relationship to the scale in Fig. 2F, are unknown, it seems like the Fig. 2G scale has an inverse relationship to stimulus complexity, in contrast to the expected positive relationship for Fig. 2F. This is presumably what creates the observed negative correlation in Fig. 2G.

Taken together, points 1-3 suggest that VHpresent and VHabsent are complex, unnecessary, and disconnected metrics for understanding target detection response times. The standard, simple explanation should stand. Task difficulty and response time in target detection tasks, in both present and absent trials, are positively correlated with target-distractor similarity.

Respectfully, we disagree with your assessment. Your last point is not logically consistent though: response times for target-absent trials cannot be correlated with any target-distractor similarity since there is no target in the first place in a target-absent array. We have shown that target-absent response times are in fact, independent of experimental context, which means that they index an image property that is independent of any reference target (Results, p. 15; Section S4). This property is what we define as visual homogeneity.

I think my interpretations apply to Experiments 3 and 4 as well, although I find the analysis in Fig. 4 especially hard to understand. The VH space in this case is based on Experiment 3 oddball detection in a stimulus set that included both symmetric and asymmetric objects. However, the response times for a very different task in Experiment 4, a symmetric/asymmetric judgment, are plotted against the axes derived from Experiment 3 (Fig. 4F and 4G). It is not clear to me why a measure based on oddball detection that requires no use of symmetry information should be predictive of within-stimulus symmetry detection response times. If it is, that requires a theoretical explanation not provided here.

We are using an oddball detection task to estimate perceptual dissimilarity between objects, and construct the underlying perceptual representation of both symmetric and asymmetric objects. This enabled us to then ask if some distance-to-center computation can explain response times in a symmetry detection task, and obtain an answer in the affirmative. We have reworked the text to make this clear.

(4) Contrary to the VH theory, same/different tasks are unlikely to depend on a decision boundary in the middle of a similarity or homogeneity continuum.

We have provided empirical proof for our claims, by showing that target-present response times in a visual search task are correlated with “different” responses in the same-different task, and that target-absent response times in the visual search task are correlated with “same” responses in the same-different task (Section S3).

The authors interpret the inverse relationship of response times with VHpresent and VHabsent, described above, as evidence for their theory. They hypothesize, in Fig. 1G, that VHpresent and VHabsent occupy a single scale, with maximum VHpresent falling at the same point as minimum VHabsent. This is not borne out by their analysis, since the VHpresent and VHabsent value scales are mainly overlapping, not only in Experiments 1 and 2 but also in Experiments 3 and 4. The authors dismiss this problem by saying that

their analyses are a first pass that will require future refinement. Instead, the failure to conform to this basic part of the theory should be a red flag calling for revision of the theory.

We respectfully disagree – by no means did we dismiss this problem! In fact, we have explicitly acknowledged this by saying that VH does not explain all the variance in the response times, but nonetheless explains substantial variance and might form the basis for an initial guess or a fast response. The remaining variance might be explained by processes that involve more direct scrutiny. Please see Results, page 10 & 22.

The reason for this single scale is that the authors think of target detection as a boundary decision task, along a single scale, with a decision boundary somewhere in the middle, separating present and absent. This model makes sense for decision dimensions or spaces where there are two categories (right/left motion; cats vs. dogs), separated by an inherent boundary (equal left/right motion; training-defined cat/dog boundary). In these cases, there is less information near the boundary, leading to reduced speed/accuracy and producing a pattern like that shown in Fig. 1G.

The key conceptual advance of our study is that we show that even target/present, same/different or symmetry judgements can be fit into the standard decision-making framework.

This logic does not hold for target detection tasks. There is no inherent middle point boundary between target present and target absent. Instead, in both types of trials, maximum information is present when the target and distractors are most dissimilar, and minimum information is present when the target and distractors are most similar. The point of greatest similarity occurs at the limit of any metric for similarity. Correspondingly, there is no middle point dip in information that would produce greater difficulty and higher response times. Instead, task difficulty and response times increase monotonically with the similarity between targets and distractors, for both target present and target absent decisions. Thus, in Figs. 2F and 2G, response times appear to be highest for animals, which share the largest numbers of closely similar distractors.

Unfortunately, your logic does not boil down to any quantitative account, since you are using vague terms like “maximum information”. Further, any argument based solely on item similarity to explain visual search or symmetry responses cannot explain systematic variations observed for target-absent arrays and for symmetric objects, for the reasons below.

If target-distractor dissimilarity were the sole driver of response times, target-absent judgements should always take the longest time since the target and distractor have zero similarity, with no variation from one image to another. This account does not explain why target-absent response times vary so systematically.

Similarly, if symmetry judgements are solely based on comparing the dissimilarity between two halves of an object, there should be no variation in the response times of symmetric objects since the dissimilarity between their two halves is zero. However we do see systematic variation in the response times to symmetric objects.

DEFINITION OF AREA VH USING fMRI

(1) The area VH boundaries from different experiments are nearly completely non-overlapping.

In line with their theory that VH is a single continuum with a decision boundary somewhere in the middle, the authors use fMRI searchlight to find an area whose responses positively correlate with homogeneity, as calculated across all of their target present and target absent arrays. They report VH-correlated activity in regions anterior to LO. However, the VH defined by symmetry Experiments 3 and 4 (VHsymmetry) is substantially anterior to LO, while the VH defined by target detection Experiments 1 and 2 (VHdetection) is almost immediately adjacent to LO. Fig. S13 shows that VHsymmetry and VHdetection are nearly non-overlapping. This is a fundamental problem with the claim of discovering a new area that represents a new quantity that explains response times across multiple visual tasks. In addition, it is hard to understand why VHsymmetry does not show up in a straightforward subtraction between symmetric and asymmetric objects, which should show a clear difference in homogeneity. • Actually VHsymmetry is apparent even in a simple subtraction between symmetric and asymmetric objects (Figure S10). The VH regions identified using the visual search task and symmetry task have a partial overlap, not zero overlap as you are incorrectly claiming.

We have noted that it is not straightforward to interpret the overlap, since there are many confounding factors. One reason could simply be that the stimuli in the symmetry task were presented at fixation, whereas the visual search arrays contained items exclusively in the periphery. Another that the participants in the two tasks were completely different, and the lack of overlap is simply due to inter-individual variability. Testing the same participants in two tasks using similar stimuli would be ideal but this is outside the scope of this study. We have acknowledged these issues in the Results (p. 26) and in the Supplementary Material (Section S8).

(2) It is hard to understand how neural responses can be correlated with both VHpresent and VHabsent.

The main paper results for VHdetection are based on both target-present and target-absent trials, considered together. It is hard to interpret the observed correlations, since the VHpresent and VHabsent metrics are calculated in such different ways and have opposite correlations with target similarity, task difficulty, and response times (see above). It may be that one or the other dominates the observed correlations. It would be clarifying to analyze correlations for target-present and target-absent trials separately, to see if they are both positive and correlated with each other.

Thanks. The positive correlation between VH and neural response holds even when we do the analysis separately for target-present and -absent searches (correlation between neural response in VH region and visual homogeneity ($n = 32$, $r = 0.66$, $p < 0.0005$ for target-present searches & $n = 32$, $r = 0.56$, $p < 0.005$ for target-absent searches).

(3) The definition of the boundaries and purpose of a new visual area in the brain requires circumspection, abundant and convergent evidence, and careful controls.

Even if the VH metric, as defined and calculated by the authors here, is a meaningful quantity, it is a bold claim that a large cortical area just anterior to LO is devoted to calculating this metric as its major task. Vision involves much more than target detection and symmetry detection. The cortex anterior to LO is bound to perform a much wider range of visual functionalities. If the reported correlations can be clarified and supported, it would be more circumspect to treat them as one byproduct of unknown visual processing in the cortex anterior to LO, rather than treating them as the defining purpose for a large area of the visual cortex.

We totally agree with you that reporting a new brain region would require careful interpretation and abundant and converging evidence. However, this requires many studies worth of work, and historically category-selective regions like the FFA have achieved consensus only after they were replicated and confirmed across many studies. We believe our proposal for the computation of a quantity like visual homogeneity is conceptually novel, and our study represents a first step that provides some converging evidence (through replicable results across different experiments) for such a region. We have reworked our manuscript to make this point clearer (Discussion, p 32).

Reviewer #2 (Public Review):

Summary:

This study proposes visual homogeneity as a novel visual property that enables observers perform to several seemingly disparate visual tasks, such as finding an odd item, deciding if two items are the same, or judging if an object is symmetric. In Experiment 1, the reaction times on several objects were measured in human subjects. In Experiment 2, the visual homogeneity of each object was calculated based on the reaction time data. The visual homogeneity scores predicted reaction times. This value was also correlated with the BOLD signals in a specific region anterior to LO. Similar methods were used to analyze reaction time and fMRI data in a symmetry detection task. It is concluded that visual homogeneity is an important feature that enables observers to solve these two tasks.

Strengths:

- (1) The writing is very clear. The presentation of the study is informative.*
- (2) This study includes several behavioral and fMRI experiments. I appreciate the scientific rigor of the authors.*

We are grateful to you for your balanced assessment and constructive comments.

Weaknesses:

- (1) My main concern with this paper is the way visual homogeneity is computed. On page 10, lines 188-192, it says: "we then asked if there is any point in this multidimensional representation such that distances from this point to the target-present and target-absent response vectors can accurately predict the target-present and target-absent response times with a positive and negative correlation respectively (see Methods)". This is also true for the symmetry detection task. If I understand correctly, the reference point in this perceptual space was found by deliberating satisfying the negative and positive correlations in response times. And then on page 10, lines 200-205, it shows that the positive and negative correlations actually exist. This logic is confusing. The positive and negative correlations emerge only because this method is optimized to do so. It seems more reasonable to identify the reference point of this perceptual space independently, without using the reaction time data. Otherwise, the inference process sounds circular. A simple way is to just use the mean point of all objects in Exp 1, without any optimization towards reaction time data.*

We disagree with you since the same logic applies to any curve-fitting procedure. When we fit data to a straight line, we are finding the slope and intercept that minimizes the error between the data and the straight line, but we would hardly consider the process circular when a good fit is achieved – in fact we take it as a confirmation that the data can be fit linearly. In the same vein, we would not have observed a good fit to the data, if there did not

exist any good reference point relative to which the distances of the target-present and target-absent search arrays predicted these response times.

In Section S1, we have already reported that the visual homogeneity estimates for each object is strongly correlated with the average distance of each object to all other objects ($r = 0.84$, $p < 0.0005$, Figure S1). Second, to confirm that the results we obtained are not due to overfitting, we have already reported a cross-validation analysis, where we removed all searches involving a particular image and predicted these response times using visual homogeneity. This too revealed a significant model correlation confirming that our results are not due to overfitting.

(2) On page 11, lines 214-221. It says: "these findings are non-trivial for several reasons". However, the first reason is confusing. It is unclear to me why "it suggests that there are highly specific computations that can be performed on perceptual space to solve oddball tasks". In fact, these two sentences provide no specific explanation for the results.

We have now revised the text to make it clearer (Results, p. 11).

(3) The second reason is interesting. Reaction times in target-present trials can be easily explained by target-distractor similarity. But why does reaction time vary substantially across target-absent stimuli? One possible explanation is that the objects that are distant from the feature distribution elicit shorter reaction times. Here, all objects constitute a statistical distribution in the feature (perceptual) space. There is certainly a mean of this distribution. Some objects look like outliers and these outliers elicit shorter reaction times in the target-absent trials because outlier detection is very salient.

One might argue that the above account is merely a rephrasing of the idea of visual homogeneity proposed in this study. If so, feature saliency is not a new account. In other words, the idea of visual homogeneity is another way of reiterating the old feature saliency theory.

Thank you for this interesting point. We don't necessarily see a contradiction. However, we are proposing a quantitative decision variable that the brain could be using to make target present/absent judgements.

(4) One way to reject the feature saliency theory is to compare the reaction times of the objects that are very different from other objects (i.e., no surrounding objects in the perceptual space, e.g., the wheel in the lower right corner of Fig. 2B) with the objects that are surrounded by several similar objects (e.g., the horse in the upper part of Fig. 2B). Also, please choose the two objects with similar distance from the reference point. I predict that the latter will elicit longer reaction times because they can be easily confounded by surrounding similar objects (i.e., four-legged horses can be easily confounded by four-legged dogs). If the density of object distribution per se influences the visual homogeneity score, I would say that the "visual homogeneity" is essentially another way of describing the distributional density of the perceptual space.

We agree with you, and we have indeed found that visual homogeneity estimates from our model are highly correlated with the average distance of an object relative to all other objects. However, we performed several additional experiments to elucidate the nature of target-absent response times. We find that they are unaffected by whether these searches are performed in the midst of similar or dissimilar objects (Section S4, Experiment S6), and even when the same searches are performed among nearby sets of objects with completely uncorrelated average distances (Section S4, Experiment S7). We have now reworked the text to make this clearer.

(5) *The searchlight analysis looks strange to me. One can easily perform a parametric modulation by setting visual homogeneity as the trial-by-trial parametric modulator and reaction times as a covariate. This parametric modulation produces a brain map with the correlation of every voxel in the brain. On page 17 lines 340-343, it is unclear to me what the "mean activation" is.*

We have done something similar. For each region we took the mean activation at each voxel as the average activation 3x3x3 voxel neighborhood in the brain, and took its correlation with visual homogeneity. We have now reworked this to make it clearer (Results, p. 16).

Minor points

(1) *In the intro, it says: "using simple neural rules..." actually it is very confusing what "neural rules" are here. Better to change it to "computational principles" or "neural network models"??*

We have now replaced this with "using well-known principles governing multiple object representations".

(2) *In the intro, it says: "while machine vision algorithms are extremely successful in solving feature-based tasks like object categorization (Serre, 2019), they struggle to solve these generic tasks (Kim et al., 2018; Ricci et al. 2021). These are not generic tasks. They are just a specific type of visual task-judging relationship between multiple objects. Moreover, a large number of studies in machine vision have shown that DNNs are capable of solving these tasks and even more difficult tasks. Two survey papers are listed here.*

Wu, Q., Teney, D., Wang, P., Shen, C., Dick, A., & Van Den Hengel, A. (2017). Visual question answering: A survey of methods and datasets. *Computer Vision and Image Understanding*, 163, 21-40.

Małkiński, M., & Mańdziuk, J. (2022). Deep Learning Methods for Abstract Visual Reasoning: A Survey on Raven's Progressive Matrices. *arXiv preprint arXiv:2201.12382*.

Thank you for sharing these references. In fact, a recent study has shown that specific deep networks can indeed solve the same-different task (Tartaglini et al, 2023). However our broader point remains that the same-different or other such visual tasks are non-trivial for machine vision algorithms.

Reviewer #1 (Recommendations For The Authors):

Nothing to add to the public review. If my concerns turn out to be invalid, I apologize and will happily accept correction. If they are valid, I hope they will point toward a new version of this paper that optimizes the insights to be gained from this impressive dataset.

Reviewer #2 (Recommendations For The Authors):

My suggestions are as follows:

(1) *Analyze the fMRI data using the parametric modulation approach first at the single-subject level and then perform group analysis.*

To clarify, we have obtained image-level activations from each subject, and used it for all our analyses.

(2) Think about a way to redefine visual homogeneity from a purely image-computable approach. In other words, visual homogeneity should be first defined as an image feature that is independent of any empirical response data. And then use the visual homogeneity scores to predict reaction times.

While we understand what you mean, any image-computable representation such as from a deep network may carry its own biases and may not be an accurate representation of the underlying object representation. By contrast, neural dissimilarities in the visual cortex are strongly predictive of visual search oddball response times. That is why we used visual search oddball response times as a proxy for the underlying neural representation, and then asked whether some decision variable can be derived from this representation to explain both target present and absent judgements in visual search.

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